INCH-POUND
MIL-DTL-32674
24 October 2020
SUPERSEDING
(See 6.7)

DETAIL SPECIFICATION

GROUND ILLUMINATION HAND HELD SIGNAL SERIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for Ground Illumination Hand Held Signal Series. The information within is intended for compliance. Requirements and verifications in both the main portion of the specification and the applicable appendix should be met for compliance.

1.2 Classification. Signals are of the following types, as specified (see 6.2).

1.2.1 Types. The types of signals are as follow:

Cluster Signals:	Green Star Cluster, M125A1
Appendix A	Red Star Cluster, M158
	White Star Cluster, M159
Parachute Signals:	Green Star Parachute, M195
Appendix B	Red Star Parachute, M126A1
	White Star Parachute, M127A1
Smoke Signals:	Green Smoke Parachute, M128A1
Appendix C	Red Smoke Parachute, M129A1
	Yellow Smoke Parachute, M194

Comments, suggestions, or questions on this document should be addressed to the U.S. Army CCDC AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000 or emailed usarm.pica.ardec.list.aredc-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-P-223	-	Powder, Black
MIL-W-12332	-	Welding, Resistance, Spot, Seam, and Projection; for Fabricating Assemblies of Low-Carbon Steel
MIL-S-23389	-	Shipping and Storage Container, Cartridge, 20MM, M548
MIL-DTL-70759	-	Container, Ammunition: Hand-Held Pyrotechnic Signal Series, PA1421A1

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available on line at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, Drawings, and publications. The following other Government documents, Drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (CCDC AC)

8797921	-	Tube, Casing
8797922	-	Seal, Rocket, Barrel
8797923	-	Seal, Cork

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8797927	-	Washer, Retaining
8797928	-	Bolt
8797929	-	Barrel, Rocket
8797936	-	Spacer, Propellant
8797937	-	Washer, Barrel Rocket
8797938	-	Washer and Disc Assembly
8797941	-	Protector Assembly
8797947	-	Tail Assembly
8797951	-	Vane, Tail
8797958	-	Washer, disc Assembly
8797961	-	Signal Body & Delay Assembly
8797963	-	Signal Body
8797980	-	Disc, Case
8797996	-	Signals, Smoke, Ground, Parachute Green Smoke, M128A1 and Red Smoke, M129A1 Signal Assembly
8887529	-	Grain, Propellant
8887530	-	Propellant Assembly
9235026	-	Plate, Exhaust
9251411	-	Housing, Delay
9251412	-	Delay Assembly
9255782	-	Signal, Smoke, Ground, Yellow, Parachute, M194
12900001	-	Packing and Marking for container, PA142 for Signal, Ground
12900009	-	Packing and Marking for container, Metal M548 for Signals, Ground13061289 - Signals Illumination, Ground Star Cluster, Ml25Al, M158, M159 Assemblies
13061289	-	Signal, Illumination Ground Star Cluster, M125A1, M158, M159 Assemblies
13061290	-	Signal, Illumination Ground Parachute, Red Star, M126A1, White Star, M127A1 and Green Star, M195 Signal Assemblies
13061297	-	Plastic Firing Cap Assembly
13061298	-	Pin, Firing

(Copies of these Drawings are available online from usarmy.pica.ccdc-ac.list.Drawing-request-help-desk@mail.mil or from US Army CCDC AC, ATTN: FCDD-ACE-SP, Picatinny, NJ 07806-5000CCDC AC)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116/108C - Palletization / Unitization for M548

(Copies of these documents may be requested online at https://www.dau.edu/cop/ ammo/ Pages/ Default.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E 11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E 70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode

(Copies of this document are available from https://www.astm.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals with all requirements specified in applicable specifications.

3.2.1 Firing pin. The firing pin shall be in accordance with Drawing 13061298.

3.2.2 Firing cap. The firing cap shall be in accordance with Drawing 13061297.

3.2.2.1 Push test.

a. The firing cap shall support its own weight in the inverted position when placed 1.125 inches onto a 1.545 inch diameter steel rod.

b. The firing cap shall completely seat on a 1.560 inch diameter steel rod when initially placed 1.125 inches onto the rod and impacted with a 1.75 inch diameter steel ball (0.789 lbs) from a drop height of 14 inches.

3.2.3 Delay housing. The delay housing shall be in accordance with Drawing 9251411.

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3.2.4 Delay assembly. The delay shall be assembled in accordance with Drawing 9251412.

3.2.5 Signal body. The signal body shall be in accordance with Drawing 8797963.

3.2.6 Signal body & delay assembly. The signal body and delay shall be assembled in accordance with Drawing 8797961.

3.2.7 Propellant grain. The propellant grain shall be in accordance with Drawing 8887529.

3.2.7.1 Overall density. The density of a propellant grain shall be 1.82 – 1.89 grams/cc.

3.2.7.2 Sectional density. The sectional density (pellet grain) at any section shall be 1.80 to 1.90 grams/cc.

3.2.7.3 Ballistic test. Utilizing an unslotted spacer, Drawing 8797936, in both the front and rear end of the propellant assembly, the signals are to be fully loaded and conditioned at hot temperature (68°C to 74°C or 155°F to 165°F). When functioned, the signals shall comply with 3.3.1 (no premature bursts, no premature ejections, and no launcher delay).

3.2.8 Propellant assembly. The propellant shall be in accordance with Drawing 8887530.

3.2.9 Exhaust plate. The exhaust plate shall be in accordance with Drawing 9235026.

3.2.10 Bolt. The bolt shall be in accordance with Drawing 8797928.

3.2.10.1 Bolt pull strength. The bolt shall meet a 4200lb minimum (min.) pull requirement. (Ultimate tensile strength of approximately 160,000 psi.)

3.2.11 Tail assembly. The tail shall be in accordance with Drawing 8797947.

3.2.11.1 Mechanical properties. The tail vane shall have the following mechanical properties in accordance with Drawing 8797951:

a. Yield strength: 170,000-220,000 psi.

b. Tensile strength: 200,000 – 240,000 psi.

c. Elongation: 3% minimum.

3.2.11.2 Welds. The welds shall be in accordance with specification MIL-W-12332 except that the minimum button diameter requirement for the peel test shall be 0.06 inches.

3.2.12 Casing tube. The casing tube shall be in accordance with Drawing 8797921.

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3.2.12.1 Hydrostatic pressure. The casing tube shall withstand 1500 psi. for a minimum of 5 seconds without deforming or rupturing.

3.2.13 Protector assembly. The protector shall be in accordance with Drawing 8797941.

3.2.14 Rocket barrel. The rocket barrel shall be in accordance with Drawing 8797929.

3.2.15 Assembly. The assembly shall be in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals.

3.2.16 Ground illumination signal. The ground illumination signal shall be in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals.

3.2.17 Black powder propellant grain. The propellant (prior to pressing) shall be in accordance with Drawing 8887529.

3.2.18 Moisture content. Moisture content shall be in accordance with Table I.

TABLE I. Moisture content.

Material	Drawing Requirement
Delay composition	9251412
Black powder	13061289
	13061290
Cork	8797922
	8797923
Felt	8797937
	8797958
Paper	8797936
	8797938
	8797958
Chipboard	8797927
	8797980
Hardboard	8797938
Wood	8797941

3.3 Functioning (all temperatures). The signal shall function and comply with the following at each of the temperatures in 3.3.1, 3.3.2, and 3.3.3.

a. Incorrect color. The color emitted from all signals/payloads when functioned shall match the label on the signal.
b. Signal fails to function. The signal shall launch, eject its payload at a minimum altitude, and the pyrotechnic (smoke/light) needs to produce its effect for a

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minimum burn time when struck by the firing cap on the first attempt. If a signal does not function on the first strike, a second attempt is allowed if there is no visible strike mark on the primer.

c. Cork sealing disc. The cork sealing disc in the end of the barrel shall not be loose.

d. Premature bursts. The illuminant or payload assembly shall not eject, ignite, and burn within a horizontal, vertical, and radial (slant) distance one-hundred (100) feet of the launcher.

e. Premature ejection. The illuminant or payload assembly shall not eject at an altitude of less than one-hundred (100) feet from the ground within a horizontal distance of two-hundred fifty (250) feet from the launcher, or at an altitude of less than fifty (50) feet from the ground at a horizontal distance in excess of two-hundred fifty (250) feet from the launcher.

f. Hang fire during functioning; Launcher delay. The signal shall completely exit the rocket barrel within two (2) seconds of striking the primer.

g. Parachute. The parachute shall not separate from the assembly, delay opening from time of ejection for more than 5 seconds, or remain partially open during ambient temperature testing. The parachute shall not separate from the illuminant assembly (Parachute opening delay shall be recorded for informational purposes as well as whether the chute remains partially open) during cold temperature testing.

h. Burning time of illuminant charge. The M125A1, M158 & M159 signals shall burn in 8.5 seconds ± 2 seconds, except that the extinction interval between the third star and the fifth star shall not exceed 4 seconds.

3.3.1 Hot temperature (155°F to 165°F). The signal shall function and comply with 3.3 when stored and functioned at 155°F to 165°F.

3.3.2 Ambient temperature (65°F to 75°F). The signal shall function and comply with 3.3 and the following when stored and functioned at 65°F to 75°F:

a. Angle. The illuminant or payload assembly shall not eject at an angle of more than 30 degrees from the vertical.

b. First trace of light or smoke. The illuminant assembly shall burn for not less than 50.0 seconds in the M126A1 and the M195 assemblies and for not less than 25.0 seconds in the M127A1 assembly. The smoke assembly shall start to emit smoke not more than 4 seconds after ejection from the signal. The burning time of the smoke assembly shall be not less than 6.0 seconds nor more than 18.0 seconds for the M128A1 and M129A1, and not less than 9.0 seconds nor more than 18.0 seconds for the M194. The burning time of the cluster assembly shall be 8.5 seconds ± 2 seconds.

c. Average altitude. The average altitude for all illuminant or payload assemblies which function above 500 feet shall be not less than 725 feet.

d. Average angle. The average angle from the vertical for all illuminant or payload assemblies which function within the maximum (max) angle from the vertical (30 degrees) shall be not more than 12 degrees.

3.3.3 Cold temperature (-70°F to -60°F). The signal shall function and comply with 3.3 and the following when stored and functioned at -70°F to -60°F:

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a. Average angle. The average angle from the vertical of the ejected illuminant or payload assembles shall be not more than 25 degrees.

3.4 Workmanship. The signals and their components and assemblies shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material.

3.5 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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4. VERIFICATION

TABLE II. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration (end item test) 3 - Examination 4 - Test		Classes of Verification A - First Article Inspection B – Conformance Inspection						
Description	Section 3 Requirements	Verification Method				Verification Class		Section 4 Verifications
		1	2	3	4	A	B	
First article	3.1.1	X	X	X	X	X		4.2
Conformance	3.1.2	X	X	X	X		X	4.3
Firing pin	3.2.1			X		X	X	4.3.2.1
Firing cap	3.2.2			X	X	X	X	4.3.2.2 & 4.5.6
Push test	3.2.2.1				X	X	X	4.3.2.2 & 4.5.6
Delay housing	3.2.3			X		X	X	4.3.2.3
Delay assembly	3.2.4			X		X	X	4.3.2.4
Signal body	3.2.5			X		X	X	4.3.2.5
Signal body & delay assembly	3.2.6			X		X	X	4.3.2.6
Propellant grain	3.2.7			X	X	X	X	4.3.2.7 & 4.5.2.1 & 4.5.2.2 & 4.5.8
Overall density	3.2.7.1	X			X	X	X	4.3.2.7 & 4.5.2.1
Sectional density	3.2.7.2	X			X	X	X	4.3.2.7 & 4.5.2.2
Ballistic test	3.2.7.3		X			X	X	4.3.2.7 & 4.5.8
Propellant assembly	3.2.8			X		X	X	4.3.2.8
Exhaust plate	3.2.9			X		X	X	4.3.2.9
Bolt	3.2.10	X		X	X	X	X	4.3.2.10 & 4.5.4
Bolt pull strength	3.2.10.1				X	X	X	4.3.2.10 & 4.5.4
Tail assembly	3.2.11	X		X	X	X	X	4.3.2.11 & 4.5.5 & 4.5.7

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TABLE II. Requirement/verification cross reference matrix – Continued.

Description	Section 3 Requirements	Verification Method				Verification Class		Section 4 Verifications
		1	2	3	4	A	B	
Mechanical properties	3.2.11.1	X			X	X	X	4.3.2.11 & 4.5.5
Welds	3.2.11.2				X	X	X	4.3.2.11 & 4.5.7
Casing tube	3.2.12			X	X	X	X	4.3.2.12 & 4.5.3
Hydrostatic pressure	3.2.12.1				X	X	X	4.3.2.12 & 4.5.3
Protector assembly	3.2.13			X		X	X	4.3.2.13
Rocket barrel	3.2.14			X		X	X	4.3.2.14
Assembly	3.2.15			X		X	X	4.3.2.15 thru 4.3.2.19
Ground illumination signal	3.2.16			X		X	X	4.3.2.20
Black powder propellant grain	3.2.17	X	X	X	X	X	X	4.3.2.21 & 4.5.9
Moisture content	3.2.18	X		X	X	X	X	4.3.2.22 & 4.5.1
Functioning – all temperatures	3.3		X	X	X	X	X	4.4
Functioning – hot	3.3.1		X	X	X	X	X	4.4.1
Functioning – ambient	3.3.2		X	X	X	X	X	4.4.2
Functioning – cold	3.3.3		X	X	X	X	X	4.4.3
Workmanship	3.4	X		X	X	X	X	4.3.2.1 – 4.3.2.26
Ammunition lot numbering	3.5			X		X	X	4.3.1
PA142 plastic container	5.1.1			X		X	X	4.3.2.27
M548 metal storage container	5.1.2			X		X	X	4.3.2.28
PA142 initial packing	5.1.3			X		X	X	4.3.2.23
PA142 final packing	5.1.4			X	X	X	X	4.3.2.24
Leakage test	5.1.4.1			X	X	X	X	4.3.2.24 & 4.5.10

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M548 packing	5.1.5			X	X	X	X	4.3.2.25
Airtightness	5.1.5.1			X	X	X	X	4.3.2.25 & 4.5.11
M548 Palletization / Unitization	5.1.5.2			X			X	4.3.2.26

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2)
b. Conformance Inspection (see 4.3)

4.2 First article inspection. When specified, a sample and its components and sub-assemblies shall be subjected to first article inspection in accordance with Table III.

4.2.1 Inspections to be performed. The first article inspection shall be 100% examination in accordance with Table III. All components, in addition to table III, shall be inspected 100% for conformance criticals.

4.2.2 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table III are not conforming.

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TABLE III. First article inspection.

Examination or test	No. of sample units	Requirement paragraph	Paragraph reference/ Inspection method
Firing pin			
Examination for defects	25	3.2.1	4.3.2.1
Firing cap			
Examination for defects	25	3.2.2	4.3.2.2
Push test	5/	3.2.2.1	4.3.2.2 Major 107 and 4.5.6
Delay housing			
Examination for defects	25	3.2.3	4.3.2.3
Delay assembly			
Examination for defects	25	3.2.4	4.3.2.4
Signal body			
Examination for defects	25	3.2.5	4.3.2.5
Signal body & delay assembly			
Examination for defects	25	3.2.6	4.3.2.6
Propellant grain			
Examination for defects	25	3.2.7	4.3.2.7
Overall density test	100%	3.2.7.1	4.3.2.7 Critical 2 and 4.5.2.1
Sectional density test	40	3.2.7.2	4.3.2.7 Critical 3 and 4.5.2.2
Propellant ballistic test	1/	3.2.7.3	4.3.2.7 Critical 4 and 4.5.8
Propellant assembly			
Examination for defects	25	3.2.8	4.3.2.8
Exhaust plate			
Examination for defects	25	3.2.9	4.3.2.9
Bolt			
Examination for defects	25	3.2.10	4.3.2.10
Bolt pull strength test	20	3.2.10.1	4.3.2.10 Major 109 and 4.5.4

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TABLE III. First article inspection - Continued.

Examination or test	No. of sample units	Requirement paragraph	Paragraph reference/ Inspection method
Tail assembly			
Examination for defects	25	3.2.11	4.3.2.11
Weld peel test	2/	3.2.11.2	4.3.2.11 Major 108 and 4.5.7
Mechanical properties test	5	3.2.11.1	4.3.2.11 Major 109 and 4.5.5
Casing tube			
Examination for defects	25	3.2.12	4.3.2.12
Hydrostatic pressure test	3/	3.2.12.1	4.3.2.12 Critical 1 and 4.5.3
Protector assembly			
Examination for defects	25	3.2.13	4.3.2.13
Rocket barrel			
Examination for defects	25	3.2.14	4.3.2.14
Assembly			
Examination for defects	25	3.2.15	4.3.2.15 thru 4.3.2.19
Ground illumination signal			
Examination for defect	25	3.2.16	4.3.2.20
Black powder propellant grain			
Examination for defects	6/	3.2.17	4.3.2.21 & 4.5.9
Moisture content			
Test for moisture	10/	3.2.18	4.5.1
PA142 Plastic container			
First article inspection	7/	5.1.1	4.3.2.27
M548 Metal storage container			
First article inspection	8/	5.1.2	4.3.2.28
PA142 Initial packing			
Examination for defects	25	5.1.3	4.3.2.23
PA142 Final packing			
Examination for defects	25	5.1.4	4.3.2.24
Leakage test	80	5.1.4.1	4.3.2.24 Major 102

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TABLE III. First article inspection - Continued.

Examination or test	No. of sample units	Requirement paragraph	Paragraph reference/ Inspection method
M548 Packing			
Examination for defects	9/	5.1.5	4.3.2.25
Airtightness	4	5.1.5.1	4.3.2.25 Major 102
Functioning – all temperatures	4/	3.3	4.4
Incorrect color	4/	3.3a	4.4
Signal fails to function	4/	3.3b	4.4
Cork sealing disc	4/	3.3c	4.4
Premature bursts	4/	3.3d	4.4
Premature ejection	4/	3.3e	4.4
Launcher delay	4/	3.3f	4.4
Functioning - hot	32	3.3.1	4.4.1
Functioning - ambient	32	3.3.2	4.4.2
Angle from vertical	32	3.3.2a	4.4.2
Height of first trace of light	32	3.3.2b	4.4.2
Average altitude	32	3.3.2c	4.4.2
Average angle from vertical	32	3.3.2d	4.4.2
Functioning - cold	16	3.3.3	4.4.3
Average angle from vertical	16	3.3.3a	4.4.3

Notes:
1/ 150 grains shall be assembled and tested in fifty (50) signals; accept 0, reject 1.
2/ Twenty-five (25) upper tail ring weld and twenty-five (25) lower tail ring weld
3/ Tubing stock used to manufacture the 25 casing tubes shall be tested
4/ The defects that apply to all temperatures shall be measured at the time of each of the temperature functions
5/ Twenty-five (25) samples shall be selected; accept 0, reject 1.
6/ A sample from the same lot used in the production of the first article quantity shall be tested in accordance with Majors 101 through 112.
7/ First article inspection of the PA142 Plastic container shall be in accordance with the PA142 specification, MIL-DTL-70759.
8/ First article inspection of the M548 metal container shall be in accordance with the M548 metal container specification, MIL-S-23389.
9/ 1 is marked as a light box (w/ only 8 signals)
10/ Refer to 4.3.2.22

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4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." In addition, each inspection lot of signal assemblies shall contain:

a. Parts of one kind from one supplier.
b. Primers of one interfix lot number.
c. Black powder of one lot.
d. Signal assemblies of one designation.
e. Energetic composition ingredients from one lot.
f. Illuminant or smoke compositions produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.
g. An ammunition lot number that is in accordance with MIL-STD-1168.

4.3.2 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraph 4.3.2.1 – 4.3.2.26, the definitions of critical, major and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.
b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.
c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

4.3.2.1	Firing pin.			Drawing Number 13061298
				Next Higher Assembly 13061297
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None Defined			
Major				
101	Width at top	Level IV	3.2.1	Gage
102	Length	Level IV	3.2.1	Gage
103	Width of ball end	Level IV	3.2.1	Gage
Minor				
201	Protective coating damaged exposing base metal	Level II	3.2.1	Visual
202	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.2	Firing cap.			Drawing Number 13061297
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Inner diameter at pushout	Level IV	3.2.2	Gage
102	Inner diameter just below top	Level IV	3.2.2	Gage
103	Inner diameter at top	Level IV	3.2.2	Gage
104	Outer diameter at top	Level IV	3.2.2	Gage
105	Outer diameter at bottom	Level IV	3.2.2	Gage
106	Length	Level IV	3.2.2	Gage
107	Push test	1/	3.2.2.1	4.5.6
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
1/ 20 samples/lot; accept 1, reject 2

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4.3.2.3	Delay housing.			Drawing Number 9251411
				Next Higher Assembly 9251412
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Diameter under flange	Level IV	3.2.3	Gage
102	Small outside diameter	Level IV	3.2.3	Gage
103	Pitch diameter of thread, max	Level IV	3.2.3	Gage
104	Minor diameter of thread, max	Level IV	3.2.3	Gage
105	Diameter of delay cavity	Level IV	3.2.3	Gage
106	Depth of delay cavity	Level IV	3.2.3	Gage
Minor				
201	Diameter of flange	Level II	3.2.3	Gage
202	Length of diameter under flange	Level II	3.2.3	Gage
203	Location of flash hole	Level II	3.2.3	Gage
204	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.4	Delay assembly.			Drawing Number 9251412
				Next Higher Assembly 8797961
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Delay charge missing	100% + VL-VII	3.2.4	Visual
Major				
101	Delay composition above flush or more than max below flush	Level IV	3.2.4	Gage
102	Flash hole obstructed	Level IV	3.2.4	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

4.3.2.5	Signal body.			Drawing Number 8797963
				Next Higher Assembly 8797961
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Small inside diameter	Level IV	3.2.5	Gage
Minor				
201	Large outside diameter, max	Level II	3.2.5	Gage
202	Large inside diameter, min	Level II	3.2.5	Gage
203	Overall length	Level II	3.2.5	Gage
204	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.6	Signal body & delay assembly.			Drawing Number 8797961
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Gap between body and delay housing over max	Level IV	3.2.6	Gage
Minor				
201	Marking missing or incorrect	Level II	3.2.6	Visual
202	Designation incorrect	Level II	3.2.6	Visual
203	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.7	Propellant grain.			Drawing Number 8887529
				Next Higher Assembly 8887530
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Crack (4 power magnification)	100% + VL-VII	3.2.7	Visual
2	Overall density test	100% + VL-VII	3.2.7.1	4.5.2.1
3	Sectional density test	1/	3.2.7.2	4.5.2.2
4	Propellant ballistic test	2/	3.2.7.3	4.5.8
Major				
101	Inside diameter	Level IV	3.2.7	Gage
102	Outside diameter	Level IV	3.2.7	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
Grains shall be assembled and loaded by same manufacturing methods as intended for production
1/ Twenty-five (25) Grains (50 cylindrical sections) to be tested; accept 0, reject 1.
2/ 150 grains to be assembled and tested in fifty (50) signals; accept 0, reject 1.

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4.3.2.8	Propellant assembly.			Drawing Number 8887530
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Pettman cement coating on interior surface where not permitted	100% + VL-VII	3.2.8	Visual
2	Pettman cement coating insufficient (slight imperfections of pin hole type or those resulting from the bristles of the brush used in applying the cement, are acceptable)	100% + VL-VII	3.2.8	Visual
3	Grain broken	100% + VL-VII	3.2.8	Visual
Major				
101	Length	Level IV	3.2.8	Gage
102	Glaze not removed from interior surfaces	Level IV	3.2.8	Visual
Minor				
201	Sheath damaged	Level II	3.2.8	Visual
202	Marking missing or incorrect	Level II	3.2.8	Visual
203	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.9	Exhaust plate.			Drawing Number 9235026
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Exhaust hole(s) missing or obstructed	100% + VL-VII	3.2.9	AAIE/Gage
Major				
101	Diameter of large end of taper	Level IV	3.2.9	Gage
102	Diameter of small end of taper	Level IV	3.2.9	Gage
103	Thickness, min	Level IV	3.2.9	Gage
104	Length of flange, max	Level IV	3.2.9	Gage
105	Diameter of flange	Level IV	3.2.9	Gage
106	True position of bolt hole with diameter under flange including perpendicularity	Level IV	3.2.9	Gage
107	True position of exhaust holes with diameter under flange	Level IV	3.2.9	Gage
108	Diameter under flange, min	Level IV	3.2.9	Gage
109	Diameter of bolt, min	Level IV	3.2.9	Gage
Minor				
201	Thickness, max	Level II	3.2.9	Gage
202	Diameter of bolt hole counterbore, max	Level II	3.2.9	Gage
203	Depth of bolt hole counterbore, max	Level II	3.2.9	Gage
204	Finish is proper	Level II	3.2.9	Visual
205	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.10	Bolt.			Drawing Number 8797928
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Length, min	Level IV	3.2.10	Gage
102	Pitch diameter of thread, min	Level IV	3.2.10	Gage
103	Major diameter of thread, min	Level IV	3.2.10	Gage
104	Diameter of flash holes, min	Level IV	3.2.10	Gage
105	Length under head	Level IV	3.2.10	Gage
106	Diameter of head, min	Level IV	3.2.10	Gage
107	Concentricity of head with thread	Level IV	3.2.10	Gage
108	Flash holes obstructed or fail to intersect each other	Level IV	3.2.10	Visual
109	Bolt pull strength test	1/	3.2.10.1	4.5.4
Minor				
201	Length from head to intersecting flash hole, min	Level II	3.2.10	Gage
202	Evidence of poor workmanship	Level II	3.4	Visual
Notes: 1/ Twenty (20) samples/lot; accept 0, reject 1.				

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4.3.2.11	Tail assembly.			Drawing Number 8797947
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Tail fin missing, cracked or inverted	100% + VL-VII	3.2.11	Visual
Major				
101	Length, max	Level IV	3.2.11	Gage
102	Location of tail vanes with diameter and each other, including maximum vane extension from centerline	Level IV	3.2.11	Gage
103	Inside diameter of upper tail ring, min	Level IV	3.2.11	Gage
104	Perpendicularity of ends with diameter	Level IV	3.2.11	Gage
105	Position of rib inside diameter with upper tail ring diameter	Level IV	3.2.11	Gage
106	Weld missing	Level IV	3.2.11	Visual
107	Weld flash excessive to extent that function will be impaired	Level IV	3.2.11	Visual
108	Weld peel test	1/	3.2.11.2	4.5.7
109	Mechanical properties test (Dwg. 8797951)	2/	3.2.11.1	4.5.5
Minor				
201	Clearance between lower tail ring and tail vane	Level II	3.2.11	Gage
202	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
1/ Thirty-two (32) samples/lot, one (1) upper tail ring weld and one (1) lower tail ring weld per assembly (64 weld total); accept 0, reject 1.
2/ Five (5) tail vanes/test samples shall be selected from each heat treated batch for mechanical properties testing.

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4.3.2.12	Casing tube.			Drawing Number 8797921
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Hydrostatic pressure test	100% + VL-VII 1/	3.2.12.1	4.5.3
2	Presence of cracks	100% + VL-VII	3.2.12	Visual
Major				
101	Perpendicularity of ends	Level IV	3.2.12	Gage
102	Total length, max	Level IV	3.2.12	Gage
103	Inside diameter of ends, max	Level IV	3.2.12	Gage
104	Hydrostatic Test Stamp Missing	100% 2/	3.2.12	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
1/ Tubing stock used to manufacture the casing tube shall be 100% hydrostatically tested as specified in Para. 4.5.3. All remaining classified defects shall be checked on the finished item.
2/ Any part found without the required test stamp shall be tested to determine if it fails the critical test. If the part does fail the critical test, it shall be treated as a critical nonconformance. If the part passes the critical test, it shall be treated as a Major nonconformance.

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4.3.2.13	Protector assembly.			Drawing Number 8797941
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major 101	Outside diameter, max	Level IV	3.2.13	Gage
Minor 201	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.14	Rocket barrel.			Drawing Number 8797929
				Next Higher Assembly 13061289, 13061290, 8797996, & 9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Crack, Draw marks, inclusions, scratches more than 0.015 inches deep in sidewall	100% + VL-VII	3.2.14	Visual/Gage
Major				
101	Diameter of primer hole, max	Level IV	3.2.14	Gage
102	Concentricity of primer hole and counterbore with the inside diameter	Level IV	3.2.14	Gage
103	Inside diameter	Level IV	3.2.14	Gage
104	Depth of primer counterbore	Level IV	3.2.14	Gage
105	Outside diameter of rear end	Level IV	3.2.14	Gage
106	Length to knurl, max	Level IV	3.2.14	Gage
Minor				
201	Wall thickness, min	Level II	3.2.14	Gage
202	Thickness through bottom of charge cavity	Level II	3.2.14	Gage
203	Diameter of charge cavity	Level II	3.2.14	Gage
204	Total length	Level II	3.2.14	Gage
205	Length to top of charge cavity	Level II	3.2.14	Gage
206	Radii missing	Level II	3.2.14	Visual
207	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.15	Assembly (signal body & delay assembly after assembly of casing tube, tail assembly, propellant assembly, exhaust plate and bolt).			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Tail assembly binds on tube casing	100% + VL-VII	3.2.15	Visual/Manual 1/
2	Exhaust plate flange missing or does not engage tail assembly	100% + VL-VII	3.2.15	Visual 2/
Major				
101	Bolt or tube casing loose	Level IV	3.2.15	Manual
102	Bolt flash hole obstructed	Level IV	3.2.15	Visual
103	Propellant assembly loose	Level IV	3.2.15	Manual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
1/ The signal body shall be vertically held, forward end up, with upper tail ring assembly contacting signal body and delay assembly. If the tail assembly fails to slide down to end of tube casing by its own weight, it shall be classified as defective.
2/ When tested as above (Note 1) signal shall be classified as defective if tail assembly falls off due to absence or inadequacy of exhaust plate flange.

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4.3.2.16	Assembly (signal body & delay assembly after assembly of tail, propellant and exhaust plate assemblies and bolt, but prior to inserting illuminant/smoke washer disc assemblies and backing washer).			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Weight of expelling charge, min 1/	Level IV	3.2.15	Balance
102	Expelling charge missing	Level IV	3.2.15	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual
Notes: 1/ Expelling charge shall be weighed just prior to insertion of charge in signal body and delay assembly.				

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4.3.2.17	Assembly (prior to assembling rocket assembly, and washer and disc assembly in rocket barrel).			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Weight of initiating charge below minimum or missing 1/	100% + VL-VII	3.2.15	AAIE
2	Primer above flush	100% + VL-VII	3.2.15	Gage
Major				
101	Initiating charge over max 1/	100% + VL-VII	3.2.15	AAIE
102	Primer missing or not sealed	Level IV	3.2.15	Visual
103	Rocket barrel washer missing	Level IV	3.2.15	Visual
104	Depth of primer incorrect	Level IV	3.2.15	Gage
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

Notes:
1/ Initiating charge may be weighed just prior to insertion of charge in rocket barrel.

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4.3.2.18	Assembly (after assembling rocket assembly but prior to inserting rocket barrel seal).			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major 101	Cork Missing	100%	3.2.15	Visual
Minor 201	Tape missing, insufficient or incorrectly applied to cork seal	Level II	3.2.15	Visual
202	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.19	Assembly (prior to assembling firing cap assembly).			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Color of coating on rocket barrel seal incorrect	100% + VL-VII	3.2.15	Visual
Major				
101	Rocket barrel seal missing, or loose	100%	3.2.15	Visual
102	Thickness of lacquer coating insufficient or too thick except that pin holes and surface roughness caused by the cork surface contour are acceptable	Level IV	3.2.15	Visual
103	Rocket barrel dented	Level IV	3.2.15	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.20	Signal, illumination, ground star cluster, parachute, and smoke parachute assembly.			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Designation and nomenclature incorrect (Label Designation DWG 8797931)	100% + VL-VII	3.2.15	Visual
2	Firing cap assembled on primer end	100% + VL-VII	3.2.15	Visual
Major				
101	Total length, max	Level IV	3.2.15	Gage
102	Label missing	100%	3.2.15	Visual
103	Assembly damaged to the extent that function will be impaired	Level IV	3.2.15	Visual
Minor				
201	Marking missing or incorrect	Level II	3.2.15	Visual
202	Contents loose	Level II	3.2.15	Manual
203	Knurl not color coated	Level II	3.2.15	Visual
204	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.21	Grain propellant (prior to pressing).			Drawing Number 8887529
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Moisture content	1/	3.2.17	4.5.9.1
102	Potassium nitrate determination	1/	3.2.17	4.5.9.2
103	Calcium carbonate determination	1/	3.2.17	4.5.9.3
104	Sulfur determination	1/	3.2.17	4.5.9.4
105	Charcoal determination	1/	3.2.17	4.5.9.5
106	Ash content	1/	3.2.17	4.5.9.6
107	Specific gravity	1/	3.2.17	4.5.9.7
108	Apparent density	1/	3.2.17	4.5.9.8
109	Granulation	2/	3.2.17	4.5.9.9
110	Lead fuze burn rate	1/	3.2.17	4.5.9.10
111	Glaze finish(if applicable)	1/	3.2.17	4.5.9.11
112	pH determination	2/	3.2.17	4.5.9.12
Minor				
201	Evidence of poor workmanship	1/	3.4	Visual

Notes:

1/ The sample shall be subjected to all tests in accordance with MIL-STD-1916 or in accordance with section 4.13 of MIL-STD-1916, (Submission of individual alternate acceptance methods for one or more contractually specified sampling inspection requirements through the Government quality assurance representative (QAR) to the procuring contracting officer (PCO) for approval at any time during the contract period of performance) except that in lieu of Table IV Continuous sampling plans (MIL-STD-1916) the following continuous sampling plan shall be used: i=11, f = 1/5, and there shall be no reduced or tightened inspection level. If any sample fails to meet any requirements, the batch (contents of a single glazing barrel) represented by the sample shall be rejected and all untested batches produced since the last batch tested shall be subjected to all of the tests. These tests shall be conducted on each succeeding batch until i=11 batches are accepted. If any failures occur when f = 1/5, revert to j = 11.

2/ All batches shall be tested for pH and granulation. If any sample fails to comply with the granulation or pH requirement, the batch represented by the sample shall be rejected.

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4.3.2.22	Moisture content.			Drawing Number 13061289, 13061290, 8797996, & 9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Delay composition moisture	1/	3.2.18	4.5.1.2
102	Black powder moisture	1/	3.2.18	4.5.1.1
103	Cork moisture	1/	3.2.18	4.5.1.3
104	Felt moisture	1/	3.2.18	4.5.1.3
105	Paper moisture	1/	3.2.18	4.5.1.3
106	Chipboard moisture	1/	3.2.18	4.5.1.3
107	Hardboard moisture	1/	3.2.18	4.5.1.3
108	Wood moisture	1/	3.2.18	4.5.1.3
Minor	None defined			
Notes: 1/ A sample shall be selected and subject to test, one sample of each material of each Drawing number from each eight hours of production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excessive moisture, the remaining unloaded material and the loaded components shall be rejected. (See 6.10)				

4.3.2.23	PA142 initial packing (PA142 container, prior to sealing).	Drawing Number

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				12900001
				Next Higher Assembly 12900009
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Signal inverted or missing	Level IV	5.1.3	Visual
102	Packing component missing	Level IV	5.1.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	3.4	Visual

4.3.2.24	PA142 final packing (PA142 container, sealed).			Drawing Number 12900001
				Next Higher Assembly 12900009
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Cover embossing missing or incorrect	100% & VL-VII	5.1.4	Visual
Major				
101	Container damaged	Level IV	5.1.4	Visual
102	Leakage test	Level IV	5.1.4.1	Test / 4.5.10
103	Label missing or improperly applied	Level IV	5.1.4	Visual
104	Presence of tamper proof tape	100%	5.1.4	Visual
Minor				
201	Marking missing or incorrect	Level II	5.1.4	Visual
202	Contents loose	Level II	5.1.4	Manual
203	Evidence of poor workmanship	Level II	3.4	Visual

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4.3.2.25	M548 packing (packing and markings for container, metal M548 for signals, ground).			Drawing Number 12900009
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None Defined			
Major				
101	PA142 container quantity incorrect or improperly inserted	Level IV	5.1.5	Visual
102	Air tightness test (after sealing)	1/	5.1.5.1	Test / 4.5.11
103	Metal container damaged (dents, splits, etc.)	Level IV	5.1.5	Visual
104	Partition missing or damaged (rips, unevenly spaced, etc.)	Level IV	5.1.5	Visual
Minor				
201	Contents loose	Level II	5.1.5	Visual
202	Marking missing or incorrect	Level II	5.1.5	Visual
203	Metallic seal missing, unsealed or improperly engaged	Level II	5.1.5	Visual
204	Evidence of poor workmanship	Level II	3.4	Visual

NOTES:
Notes
1/ One (1) container out of each twenty (20) containers packed out/sealed shall be tested. If the container fails the test, it should be removed from the lot and all containers back to the last successful test should be leak tested.

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4.3.2.26	M548 Palletization / Unitization			Drawing Number 19-48-4116/108C
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None Defined			
Major				
101	Incorrect, missing, or loose strapping	100%	5.1.5.2	Visual
102	Incorrect number of containers on pallet (except for light loads)	100%	5.1.5.2	Visual
103	Containers are loaded on pallet so that there is at least one container is showing each side marking on each side of the unit load.	100%	5.1.5.2	Visual
104	Marking missing, incorrect, or illegible	Level III	5.1.5.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	5.1.5.2	Visual

4.3.2.27 PA142 plastic container. The PA142 plastic container shall be inspected in accordance with MIL-DTL-70759.

4.3.2.28 M548 metal ammunition can. The M548 metal ammunition can shall be inspected in accordance with specification MIL-S-23389.

4.4 Functioning. The signal assemblies shall be observed for any evidence of failure in accordance with Table V when sampled in accordance with Table VI, 4.4.1, 4.4.2, and 4.4.3. Video footage shall be recorded and saved to adequately capture the functioning requirements of the signals during testing. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples shall be fired within three minutes of removal from the conditioning chamber. The performance of the sample shall be observed for compliance with the requirements in table VI.

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TABLE V. Functioning defects.

Examination or Test	Category	Requirement Paragraph	Inspection Method
All temperatures			
Incorrect color	Critical	3.3a	Visual
Signal fails to function	Major	3.3b	Visual
Cork sealing disc in end of barrel is loose	Major	3.3c	Visual/Manual
Premature burst	Critical	3.3d	Gage
Premature ejection	Major	3.3e	Gage
Hangfire during functioning	Critical	3.3f	Gage
Hot temperature (155°F to 165°F)			
Ambient temperature (65°F to 75°F)			
Angle from vertical over max	Major	3.3.2a	Gage
First trace of light	Major	3.3.2b	Gage
Average altitude under min	Major	3.3.2c	Gage
Average angle from vertical	Major	3.3.2d	Gage
Cold temperature (-70°F to -60°F)			
Average angle from vertical over max	Major	3.3.3a	Gage

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TABLE VI. Functioning sampling plan.

Temp	First Article Sampling Plan		Initial Sampling Plan		Reduced Sampling Plan (After 3 Consecutive Complying Lots)	
	Number of Tests	Reject if any of the following occur	Number of Tests	Reject if a critical defect or any of the following occur	Number of Tests	Reject if a critical defect or any of the following occur
Hot	32	1 or more major or critical defects	50	1 or more major defect	20	1 or more major defect
Amb	32	1 or more major or critical defects	100	6 or more major defects or	50	4 or more major defects or
				average altitude under minimum or		average altitude under minimum or
				average angle from vertical exceeds maximum		average angle from vertical exceeds maximum
Cold	16	1 or more major or critical defects	50	4 or more major defects or	32	3 or more major defects or
				average angle from vertical exceeds maximum		average angle from vertical exceeds maximum

4.4.1 Hot temperature (155°F to 165°F). The packed signals within the plastic container shall be conditioned for a minimum of 16 hours at 155°F to 165°F. The signals shall be functioned in accordance with paragraph 4.4.

4.4.1.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table V and Table VI, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.1. The lot shall be rejected if it fails to comply with the acceptance criteria specified in Table V and Table VI.

4.4.1.2 Reduced sampling plan. After three (3) consecutive lots have complied with the criteria of 4.4.1.1, twenty (20) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table V and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, and 4.4.3.

4.4.2 Ambient temperature (65°F to 75°F). The packed signals shall be conditioned for a minimum of 16 hours at 65°F to 75°F. The signals shall be functioned in accordance with paragraph 4.4. Assemblies which function at 500 feet or below shall not be used to calculate the average altitude. Assemblies which function at 30 degrees or more shall not be used to calculate the average angle.

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4.4.2.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table V and Table VI, one hundred (100) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table V and Table VI.

4.4.2.2 Reduced sampling plan. After three consecutive lots have complied with the criteria of 4.4.2.1, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table V and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, and 4.4.3.

4.4.3 Cold temperature (-70°F to -60°F). The packed signals shall be conditioned for a minimum of 16 hours at -70°F to -60°F. The signals shall be functioned in accordance with paragraph 4.4.

4.4.3.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table V and Table VI, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.3. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table V and Table VI.

4.4.3.2 Reduced sampling plan. After three (3) consecutive lots have complied with the criteria of 4.4.3.1, thirty-two (32) assemblies shall be selected from each lot for testing and conditioned in accordance with 4.4.3. The lot shall be rejected if it fails to meet the acceptance criteria specific in Table V and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, and 4.4.3.

4.5 Test methods and procedures.

4.5.1 Moisture content

4.5.1.1 Black powder. The moisture content of the black powder shall be determined in accordance with MIL-P-223.

4.5.1.2 Delay composition. An accurately weighed portion of approximately 5 grams (g) of delay composition shall be placed in dried tared aluminum weighing dish 2 inches in diameter. A polytetraflourethylene cover shall be placed on the dish and the dish, contents and cover accurately weighed. The cover shall be removed and the dish and contents heated at 95 +/- 2°C for 2 hours min. At the end of the 2 hour drying period, remove the dish from the heat, replace the cover, and cool to room temperature in a desiccator. After cooling, accurately weigh the dish, cover and contents. The loss in weight shall be calculated as the percent of moisture using the following method:

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$$Moisture\ \% = 100 * \frac{(A - B)}{C}$$

Where:
A = weight of dish, cover and contents before drying
B = weight of dish, cover and contents after drying
C = weight of sample before drying

4.5.1.3 Felt, fiberboard, cork, wood, paper products. An accurately weighed portion of at least 10 grams of the material weighed to the nearest milligram shall be transferred to a tared weighing dish and replaced in an oven and heated at a temperature of 100 +/- 2 ℃ for 2 hours minimum. The dish and contents shall be removed from the oven and placed in a desiccator, cooled, and weighed. The difference in weight shall be calculated to percentage of moisture in the sample.

4.5.2 Density of propellant grain.

4.5.2.1 Overall density. The propellant grain shall be accurately measured, the volume determined and converted to cubic centimeters. The grain shall then be accurately weighed and the density in grams per cubic centimeter calculated and compared with the requirements. Any propellant grain that fails to comply with the requirements shall be considered defective.

4.5.2.2 Sectional density. Each of the grains shall be cut into two (2) approximately equal size cylindrical sections. Each cylinder shall be accurately weighed. Fill a beaker with clean mercury and insert the immersing fixture and level off the mercury by pressing down with the flat glass plate until it is even with the top of the beaker. Then place the cylinder beneath the prongs of the immersing fixture and level the mercury by pressing down with the flat glass until it is even with the top of the beaker. Catch the excess mercury in a clean evaporating dish. Weigh the mercury displaced by the cylinder. Calculate the density of each cylinder as follows:

Density = 13.54*W divided by the weight of the displaced mercury in grams.

Where:
W = weight of the cylinder in grams.

If any section of the grain fails to comply with the requirements, the lot shall be rejected.

4.5.3 Hydrostatic test of tube, casing. In lieu of testing each individual casing tube, the tubing stock, which is used to manufacture the casing tube, shall be 100% tested in a hydrostatic test device. 1500 psi minimum shall be applied for not less than 5 seconds. The tubing shall be observed for evidence of failure to comply with the requirements. Any full length of tubing that bursts, cracks, or shows evidence of distortion shall be considered defective.

4.5.4 Bolt pull strength. The bolt shall be tensioned under 4200 lbs. If any bolt permanently deforms, breaks, or fails to comply with the requirements on Drawing 8797928, it shall be considered defective.

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4.5.5 Mechanical properties of tail vane. The tail vane/test samples shall be tested for mechanical properties in accordance with ASTM E8/E8M. If any tail vane/test sample fails to comply with the criteria specified on the Drawing 8797951, the heat treated batch represented by the sample shall be rejected.

4.5.6 Push test of firing cap. The firing cap shall be tested in accordance with the equipment and procedure specified on Drawing 13061297. If any assembly fails to comply with the requirements of 3.2.2.1, it shall be considered defective.

4.5.7 Weld peel test. The weld on the tail assembly shall be tested in accordance with the procedures specified in MIL-W-12332. After test the assembly shall be examined to determine compliance with the Drawing requirements. If any weld fails to comply with the requirements, the lot shall be rejected.

4.5.8 Propellant ballistic test, Critical defect. Utilizing propellant spacer, Drawing 8797936, in the forward and aft end, 50 signals per every lot of loose propellant composition are to be ballistically tested in fully loaded signals at hot temperature in accordance with 4.5.2.1. Upon successfully meeting the requirements (See para. 3.3), the propellant composition used and the method of manufacture of the propellant assembly shall be considered adequate for production. All production shall then utilize the spacer, propellant Drawing 8797936 in the aft end and the slotted spacer propellant, Drawing 9328587 in the forward (delay assembly end) section of the tube, casing, and Drawing 8797921.

4.5.9 Black powder propellant grain characteristics (prior to pressing). A non-ground sample is used for all procedures except chemical composition determination. For chemical composition determination, a 30 gram portion of the sample shall be ground through a No. 60 U.S. Alternative Sieve complying with ASTM E11. All precautions shall be taken to avoid unnecessary exposure of the sample to the air; hence, as soon as a portion is ground and sifted, it shall be placed in a bottle and tightly stoppered. The grinding and sifting per each portion shall take less than 3 minutes. Additional 30 gram portions shall be ground as needed. The powdered sample shall be mixed well before analysis. The characteristics of the hand held signal propellant shall be determined as follows:

4.5.9.1 Black powder propellant moisture (prior to pressing). From the non-ground portion of the primary sample, a weighed portion of between two (2) and three (3) grams shall be transferred to a tared porcelain crucible with cover and dried between four (4) and five (5) hours at 70 + 5°C. The sample shall be covered, cooled in a desiccator (10 to 15 minutes) and weighed. The loss in weight shall be calculated in terms of percentage moisture. Three such determinations shall be made and the average of the three results reported as the percent moisture.

Procedure

1. Obtain sample of propellant composition. Keep sample sealed and let it sit in lab for a minimum of 1 hour before opening to equalize temperature.
2. Weigh a clean dry crucible with cover and record as tare weight. Make sure the crucibles and covers are marked with matching IDs and always weighed together as a

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set. Note: Always use the crucible tongs when handling crucibles and crucible covers.
3. Place between two (2) and three (3) grams of propellant composition in crucible and replace crucible cover. Record weight. Minimize the time the propellant composition is open to the air before weighing.
4. Place crucibles in 70 to 75°C oven and remove crucible covers from crucibles.
5. Keep crucibles and crucible covers in oven for four (4) to five (5) hours, with 4.5 hours being optimal.
6. At the end of the drying time, open oven and place the crucible covers back onto the matching crucibles before removing from the oven.
7. Remove crucibles from the oven and immediately place into the desiccators. The samples should be put into the desiccators directly from the oven with no intermediate handling in between.
8. Place desiccators cover onto desiccators and assure a good seal.
9. Let samples cool between 10 and 15 minutes.
10. Remove covered crucible from desiccators and immediately record weight.
11. Calculate the loss in weight in terms of percentage moisture.

4.5.9.2 Potassium nitrate composition determination. From the ground portion of the primary sample, a weighed portion of 10 ± 0.01 grams shall be recorded and transferred to a 400 ml beaker and 200 ml of distilled water added. The beaker shall be placed on a steam bath, brought to a boil, and held for 15 minutes at boiling. The contents of the beaker shall then be filtered through a tared filtering crucible and washed with successive portions of 10 to 15 ml of water. The bath water passing through the crucible shall be tested with an excess of concentrated sulfuric acid containing a few crystals of diphenylamine, until there is no blue color (a blue color indicates the presence of nitrate). Place crucible in 70 to 75ºC oven and remove cover from crucible. Keep crucible in oven for four (4) to five (5) hours. At the end of drying time, open oven and place crucible cover back onto crucible prior to removing it from the over. With the lid on, place the crucible with cover into a desiccator and seal the desiccator. Allow the sample to cool between 10 and 15 minutes. Remove the desiccator lid and immediately weigh and record the weight. The loss in weight represents moisture and potassium nitrate. The percent potassium nitrate shall be calculated on a moisture free basis, utilizing the result obtained in 4.5.9.1 as the percent moisture. The crucible and its contents shall be reserved for the determination of calcium carbonate (See 4.5.9.3)

4.5.9.3 Calcium carbonate composition determination. After determination of potassium nitrate as specified in 4.5.9.2, the residue remaining in the crucible from that procedure shall be washed with two portions of 10 to 15 ml of 10% hydrochloric acid (HCL) and one portion of 10 to 15 ml of ethanol. The crucible shall be dried for 30 minutes at 103 ± 2 ºC cooled in a desiccator and weighed. The loss in weight shall be calculated on a moisture free basis, utilizing the result obtained in 4.5.9.1 as the percent moisture. The crucible and contents shall be reserved for determination of sulfur. (See 4.5.9.4)

4.5.9.4 Sulfur composition determination. After the determination of calcium carbonate, as specified in 4.5.9.3, the crucible shall be placed in an extractor in a water bath, and extracted for 4 hours with Carbon Disulfide. After the extraction, the crucible shall be washed once with Alcohol and once with Ether, using suction, dried for one (1) hour at 100 ± 3 ºC, cooled in a

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desiccator, and weighed. The loss in weight shall be obtained by subtracting this weight from the weight of the crucible and contents, after extraction, and the result calculated to percent sulfur on a moisture free basis utilizing the result obtained in 4.5.9.1. The crucible and contents shall be reserved for the determination of charcoal. (See 4.5.9.5)

4.5.9.5 Charcoal composition determination. The residue remaining in the crucible, reserved from 4.5.9.4, shall be considered to be charcoal. Its weight shall be determined by subtracting the tare weight of the filtering crucible from the total weight of the crucible and residue. This weight shall be calculated to percent charcoal on a moisture-free basis utilizing the result in 4.5.9.1. The crucible and contents shall be reserved for ash content determination (see 4.5.9.6)

4.5.9.6 Ash composition determination. The crucible with its residue (see 4.5.9.5) shall be ignited in a muffle furnace or over a Bunsen burner until all the carbon is burned off, cooled in a desiccator and weighed. The weight of the residue shall be calculated as percent ash.

4.5.9.7 Specific gravity. A helium gas pycnometer (Micromeritics, Accupyc 1330, or equivalent) shall be used. An analytical balance accurate to ± 0.01 mg or better (Fairfax or equivalent) shall be used. The pycnometer shall be calibrated and operated at the conditions and parameters specified herein. Pycnometer manufacturer instructions and procedures shall be followed for the detailed step-by-step operation of the gas pycnometer.

4.5.9.7.1 Calibration. Calibration shall be performed using factory supplied steel balls (or steel cylinders) with certified values. For refined calibration or trouble-shooting, other certified standards (such as glass bead standards certified by a nationally recognized institute for standards) may be used. The pycnometer shall be calibrated using parameters and conditions specified below:

Helium flow pressure:	20-22 psig (recommended)
Number of purges:	5
Number of runs:	5
Purge pressure:	19.5 to 19.6 psig
Run pressure:	19.5 to 19.6 psig
Equilibration rate:	0.005 psig / min
Use run precision:	No (see note 6.11)
Equipment warm-up:	15 minutes from stand-by and 1 hour minimum from cold start-up.

4.5.9.7.2 Procedure for analysis. A sample of black powder shall be acclimated to room temperature before analysis. During winter months, let the unopened sample container warm up to room temperature for at least 16 hours prior to opening the sample container. The black powder sample shall be tested as received. Any action/operation that may cause size attrition or modification, such as grinding or sieving shall be avoided. The pycnometer shall be checked to verify that it is operating properly prior to sample analysis. An empty sample cup is weighed to the nearest 0.1 mg. Transfer a quantity of sample (about 10 grams) to the cup. The powder must be packed and the cup at least 2/3 full to obtain optimum sample weight for a high accuracy analysis. The cup and sample shall be weighed to the nearest 0.1 mg. Place the cup (sample) in

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the pycnometer chamber and lock the chamber. The sample shall be analyzed following step-by-step operating instructions from the pycnometer manufacturer and at the operating parameters and conditions specified below:

Helium flow pressure:	20-22 psig (recommended)
Number of purges:	5
Number of runs:	5
Purge pressure:	16.2 to 16.3 psig or 19.5 to 19.6 psig
Run pressure:	16.2 to 16.3 psig or 19.5 to 19.6 psig
Equilibration rate:	0.0250 psig / min
Use run precision:	Yes (see note 6.11)
Equipment warm-up:	15 minutes from stand-by and 1 hour minimum from cold start-up.

Calculate the five (5) volumes from the five (5) runs, and average the five (5) readings. Calculate the specific gravity of the sample by dividing the individual mass of the sample by its corresponding averaged volume.

4.5.9.8 Apparent density. A 50 ml graduated cylinder, having an internal diameter of approximately 1 inch, shall be weighed to the nearest 0.1g. The cylinder shall be filled with a portion of the non-ground sample as close as possibly to the 50 ml mark. The cylinder shall be suspended in an upright position by a length of cord, and allowed to fall 30 times from a height of 1 inch onto a rubber cushion. (A standard laboratory rubber stopper, having a diameter slightly larger than that of the base of the graduated cylinder, has been found satisfactory for use). Care shall be taken to prevent any oscillation or sidewise motion. Upon completion of the 30 drops, sufficient powder to raise the level to the 50 ml mark shall be added. The cylinder and contents shall be weighed to the nearest 0.1 g. The apparent density shall be calculated as follows:

$$\text{Apparent Density} = \frac{A - B}{50}$$

Where:
A = Weight of cylinder plus black powder
B = Weight of empty cylinder

4.5.9.9 Granulation. The specified sieves (see 8887529), shall be properly superimposed and assembled with a bottom pan. The sieves shall comply with ASTM E11. A weighed portion of approximately 100g of the sample shall be placed on the topmost sieve of the assembly. The assembly shall be covered and shaken for three minutes by hand or by means of a mechanical shaker geared to produce 268-315 gyrations and 150 ± 10 taps of the striker per minute. The portions retained or passed by the various sieves, as required, shall be weighed and the results calculated to a percentage basis.

4.5.9.10 Lead fuze burn rate determination. Lead tubing conforming to AECI Raw Material Specification 283B, cut into length of approximately 405 mm shall be used for this test. A pair of copper/beryllium pliers shall be used to close and round off one end of the tube. The open end of the tube shall be tapered to form a funnel-shaped cone. With the open end of the

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tube facing downwards, tap the tube to remove any lead particles from the inside of the tube. Holding the prepared lead tube vertically with the open end up, slowly fill the entire length of the tube with the propellant using a funnel. Keeping the tube in a vertical position, vibrate the tube for 30 seconds. Refill the tube to the top with a further portion of the sample. Repeat the vibrations for another 30 seconds. Insert a cotton wool plug into the open end of the tube. Using the pliers, close and round off the end of the tube. Apply oil to the rolling machine. Feed the filled tube into the rolling machine starting in the largest guide opening in the machine. Continue this process by feeding the tube through successively small guide openings being sure to rotate the tube each time so that the end of the tube which left the machine last is fed in first. Continue until the lead tube has been rolled through each guide opening in the machine. Wipe off the excess oil with a soft cloth. Each tube shall be cut to a length of one meter by cutting off approximately equal length from each end of the tube. Each tube shall be placed into the trough of a constant temperature bath at 25 ± 0.2 C for not less than two minutes. Each tube shall be lit and the burn shall be recorded to the neared 0.1 seconds.

4.5.9.11 Glaze finish (if applicable). The propellant shall be examined visually to ascertain that it has been glazed with graphite and that surfaces of the grains do not have a mottled appearance, which may indicate inadequate glazing or high moisture content.

4.5.9.12 pH determination. From each selected sample a weighed portion of 6.76 ± 0.05 g shall be placed into a 250 ml beaker. Add 100 ml of neutral distilled (or deionized) water. Cover the beaker and digest on a steam bath for 30 minutes. With stirring, cool to 25 degrees C. Determine the pH at 25 degrees C in accordance with ASTM E70 with a pH meter that is calibrated for 25 degrees C.

4.5.10 Leakage of plastic PA142 container (after sealing). The packed plastic container (Drawing 12900001) shall be placed in the two chamber leak test cylinder. Apply air in the cylinder to a pressure of three (3) to five (5) pounds per square inch (psi) and hold for eight (8) seconds minimum. Observe the pressure indicator for a drop in pressure during the holding time. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.5.11 Airtightness of metal M548 ammunition can (after sealing). The metal container shall be conditioned to ambient temperature and pressure prior to testing. The container shall be tested in accordance with 4.5.11.1 or 4.5.11.2. A wetting agent may be used to minimize air bubbles clinging to the exterior surface. A stream of recurring succession of bubbles from any surface, seam or junction shall be evidence for a defective container. Any container that fails to comply with the requirement shall be classified defective and removed from the lot and the test is to be conducted on one container out of the 20 containers that are selected for the sample.

4.5.11.1 Vacuum method. The container shall be tested for leakage by immersing the closed container in the inverted position under water in a vacuum vessel and lower the pressure in the vessel to three (3) pounds per square inch minimum below ambient pressure. Observation for leakage of air from the container interior shall be made for a minimum of three (3) seconds after reduction of pressure. The container should be completely covered by water to a level that would allow a clear view of the bubbles, at least 2 inches.

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4.5.11.2 Hot water method. The container shall be tested for leakage by immersing the closed container in the inverted position to a depth of one inch below the surface of the water. The temperature of the water and the length of time of immersion shall be that which will assure an increase in pressure to a minimum of three (3) pounds per square inch (psi) above ambient pressure. Observation for leakage of air from the container interior shall be made during the period of pressure buildup and for thirty (30) seconds after the three (3) pounds per square inch (psi) pressure differential has been reached

5. PACKAGING

5.1 Packaging requirements.

5.1.1 PA142 plastic container. The PA142 plastic container shall be in accordance with MIL-DTL-70759 and Drawing 12900001.

5.1.2 M548 metal storage container. The M548 metal storage container shall be in accordance with MIL-S-23389 and Drawing 7258943.

5.1.3 PA142 initial packing. After inserting the signal into the container, but prior to sealing, the plastic container shall be in accordance with Drawing 12900001.

5.1.4 PA142 final packing. After sealing the PA142, the container shall be in accordance with Drawing 12900001.

5.1.4.1 Leakage test. The packed PA142 shall not leak air when a differential of three (3) to five (5) psi is applied outside.

5.1.5 M548 packing. The metal M548 container shall be in accordance with Drawing 12900009.

5.1.5.1 Airtightness. The metal M548 container shall not leak when submerged under water with a three (3) psi vacuum below ambient pressure being applied.

5.1.5.2 Palletization / Unitization. The metal M548 container shall be palletized/unitized in accordance with Drawing 19-48-4116/108C.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Hand Held Signal Series consists of a firing cap that is removed from the front of the system and put onto the end, and hit to launch a rocket that rises to a height of about 850 feet where it ejects a cluster of 5 flares, or a single candle or smoke assembly attached to a parachute.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1 and 6.6).
c. Requirements for First Article samples (see 4.2 and 6.6).
d. Provision for submission of first article samples.
e. Requirements for submission of Acceptance inspection equipment (see 6.3).
f. Requirements for Ammunition Lot Numbering (see 4.3.1).
g. Critical Characteristics Clause (CCC).
h. Statistical Process Control (SPC) clause.
i. Type (see 1.2).

6.3 Submission of acceptance inspection equipment. Submit equipment designs as required to Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: RDAR-EIQ-MB, Picatinny Arsenal, NJ 07806-5000. Inspection equipment used for acceptance of product should be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% of the total tolerance of the characteristic being inspected.

6.4 Distribution of ammunition data cards. Distribution of data cards should include the following: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: RDAR-EIQ-MB, Picatinny Arsenal, NJ 07806-5000.

6.5 Justification of critical defects.

6.5.1 Delay charge missing (4.3.2.4 critical 1). If the delay charge is missing from the delay assembly, the payload or flares of the signal may eject at the same time the propellant ignites. This would eject the ignited flares or parachute with candle slightly above the head of the user and they may hit him.

6.5.2 Crack (4 power magnification) (4.3.2.7 critical 1). Cracked propellant results in a larger surface area and a much faster burn rate. This burn rate may increase to the point that the propellant can break its metal enclosure around the time of ignition and shoot out shrapnel.

6.5.3 Overall density test (4.3.2.7 critical 2). The density relates to the burn rate of the propellant. If the propellant burns to fast, the metal enclosure may burst and release shrapnel. If the propellant burns too slow, the signal will have a low flight path, possibly coming back at the user, or ejecting the flares at a height low enough for them to come down upon the user while still burning.

6.5.4 Sectional density (4.3.2.7 critical 3). If there is a large difference in the density of the different sections of the propellant than adverse things may happen. The propellant will burn at varying rates, possibly causing erratic flights where the signal may come back down upon the user or eject flares at a height low enough for them to come down upon the user while still burning.

6.5.5 Propellant ballistic test (4.3.2.7 critical 4). This test ensures the propellant does not crack under the large amount of pressure it experiences. By using to non-slotted spacers, the gas

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is not allowed to reach all places of the enclosure as fast, increasing the amount of pressure. If the propellant were to crack, there could be a catastrophic failure (see 6.6.2).

6.5.6 Pettman cement coating on interior surface where not permitted (4.3.2.8 critical 1). Pettman cement inhibits the burning of the propellant, and if it is on the interior surface, several things may occur. The propellant may burn in an adverse way and possibly crack (see 6.6.2). The propellant may also not ignite at all, while still allowing the delay to ignite. If the delay ignites, then the ignited candles will be ejected possibly directly onto the user.

6.5.7 Pettman cement coating insufficient (slight imperfections of pin hole type or those resulting from the bristles of the brush used in applying the cement are acceptable) (4.3.2.8 critical 2). If the pettman cement coating is insufficient, the propellant may burn at a rate too high, which could result in a catastrophic failure (see 6.6.2).

6.5.8 Grain broken (4.3.2.8 critical 3). If the grain is broken, there will be a larger surface area which may result in a burn rate too large which may cause a catastrophic failure (see 6.6.2).

6.5.9 Exhaust hole missing (4.3.2.9 critical 1). If the exhaust hole is missing, the propellant will burst the container it is inside releasing shrapnel (see 6.6.2) and may also eject the burning flares onto the user.

6.5.10 Tail fin missing, cracked or inverted (4.3.2.11 critical 1). If the tail fin is not present or able to function properly, the signal may have an erratic flight, possibly spinning around and striking the user, or ejecting flares near upon the user.

6.5.11 Hydrostatic pressure test (4.3.2.12 critical 1). This test checks the strength and ability of the casing tube, tube surrounding propellant, to withstand pressure. This tube undergoes a large amount of pressure and must contain the propellant, if it bursts, shrapnel will be launched out and the signal will have an erratic flight.

6.5.12 Presence of cracks (4.3.2.12 critical 2). Any cracks present on this tube may result in a weak strength of the casing tube and may cause failure (see 6.6.10).

6.5.13 Crack (4.3.2.14 critical 1). The rocket barrel is held in the users hand and contains the initial burst that ejects the hand held signal. If the rocket barrel is weak, it may burst in the user's hand.

6.5.14 Draw marks, inclusions, scratches more than 0.015 deep in sidewall (4.3.2.14 critical 2). See 6.5.12

6.5.15 Tail assembly binds on tube casing (4.3.2.15 critical 1). The tail controls the flight of the rocket, and it must slide back after the rocket is ejected to ensure a correct balance of weight. If the tail does not slide back, erratic flight will be caused which may allow the rocket to strike the user.

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6.5.16 Exhaust plate flange missing or inadequate to engage tail assembly (4.3.2.15 critical 2). If the tail of the rocket does not engage, it will not stay with and control the rocket. This will cause the rocket to have erratic flight and possibly strike the user.

6.5.17 Weight of initiating charge below min (4.3.2.17 critical 1). The initiating charge is needed to eject the rocket out and give it the initial momentum to fire vertically. If that initial momentum is too small, the rocket may wobble and fly erratically possibly striking someone.

6.5.18 Initiating charge missing (4.3.2.17 critical 2). The initiating charge is needed to eject the rocket out and give it the initial momentum to fire vertically. If that initial momentum is too small, the rocket may wobble and fly erratically possibly striking someone.

6.5.19 Primer above flush (4.3.2.17 critical 3). A primer above flush may result in inadvertent firing of the signal. This firing may cause the ejected rocket to strike someone.

6.5.20 Color coating on rocket barrel seal incorrect (4.3.3.20 critical 1). If the label on the rocket barrel seal is incorrect, the wrong signal may be sent out. These differing signals may result in a large problem of coordination on the battlefield.

6.5.21 Designation and nomenclature incorrect (4.3.2.20 critical 1). See 4.3.2.20.

6.5.22 Firing cap assembled on primer end (4.3.2.20 critical 2). If the firing cap is assembled on the primer end, the signal is in a ready to fire position and may inadvertently be launched (see 4.3.2.20).

6.5.23 Embossing. If the incorrect color is observed during functioning then there was a labeling error (see 6.5.20).

6.5.24 Premature burst during functioning. If the flares are ejected and ignited within 100 feet then they have the potential to strike the user while still ignited.

6.5.25 Color designation incorrect (A.4.3.2.1 critical 1). See 6.5.21

6.5.26 Color designation incorrect (B.4.3.2.1 critical 1). See 6.5.21

6.5.27 Color designation incorrect (C.4.3.2.1 critical 1). See 6.5.21

6.5.28 Cover embossing missing or incorrect (4.3.2.24 critical 1). This is a critical due to the labeling issue regarding these signals. Colors may be designated as a command for a mission, and an incorrect command from an incorrectly labeled signal can directly lead to casualties on the battlefield.

6.5.29 Hangfire. Signal should function within 2 seconds (>2 seconds is critical) of being initiated; see specification, section 3.3 f. If a hang fire situation occurs, it could lead to an inadvertent functioning of the signal. Inadvertent functioning of the signal may cause the signal to impact or eject its burning flares in close proximity to people or structures. If the signal

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impacts or ejects burning flares in close proximity to people and structures, it may lead to serious injury and significant damage to property.

6.6 Packaging inspections. One (1) Signal is loaded into each PA142 green plastic container. Twenty-four (24) of these loaded PA142 containers are then inserted into a single larger M548 Metal Ammunition Can (MIL-S-23389). Both the PA142 and M548 ammo can are leak tested.

6.7 Superseding documents.

MIL-S-13261G
10 January 1972,
MIL-S-13257H
12 February 1980,
MIL-S-13303G
14 September 1984.

6.8 Moisture content information. Experience has shown that inert parts placed in heat rooms at 125°F minimum for 12 hours minimum in open trays is more than sufficient to render these parts dry. After initial verification of this process, surveillance should be sufficient to maintain confidence of dry inert parts.

6.9 Definitions.

6.9.1 Analysis. The use of analytical methods and mathematical formulas to determine if an item meets a requirement.

6.9.2 Cluster signal. This refers to the M125A1 Green Cluster Signal, M158 Red Cluster Signal, or M159 White Star Cluster, which ejects a cluster of 5 flares when the delay burns through.

6.9.3 Demonstration. A functional test (for example, ballistic firing) that provides verification that a requirement is met.

6.9.4 Examination. Inspections utilizing gaging or visual means to assure conformance to requirement.

6.9.5 Flare / candle. The ejected payload, either a cluster of 5 or a single one attached to a parachute, that emits red, green, or white light.

6.9.6 Parachute signal. This refers to the M195 Green Star Parachute, M126A1 Red Star Parachute, or M127A1 White Star Parachute, which eject a single candle attached to a parachute when the delay burns through.

6.9.7 Rocket. The payload carrier that is ejected from the casing tube and ejects the payload when the delay burns through.

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6.9.8 Smoke signal. This refers to the Green Smoke Parachute, M128A1, Red Smoke Parachute, M129A1 and Yellow Smoke Parachute, M194, which eject a colored smoke that emits red, green, or yellow smoke suspended from a parachute.

6.9.9 Test. The conduct of specific procedures and methods to provide data verifying a requirement is met.

6.9.10 Premature burst. A signal which ejects, ignites, and burns at a distance less than that specified in the requirement paragraph (see 3.3d).

6.9.11 Premature ejection. A signal which in the illuminant assembly ejects but fails to ignite (see 3.3e).

6.10 Key words.

Cluster
Parachute
Smoke
Pyrotechnics

6.11 History of changes. This specification combines 3 individual specifications, MIL-S-13303 for smoke parachute signals, MIL-S-13261 for cluster signals, and MIL-S-13257 for candle parachute signals. The common parts are in the main portion of the specification and the unique parts such as the candle and parachute are included in the applicable appendix. Requirements in both the main portion of the specification and the applicable appendix must be met for compliance. The following ECP's have been incorporated into this spec: R09Y2001, R11Q2001, R16Y2001, R07Q2002, R14Y2005, R04Y2012, R06Y2012, R05Y2013, and R06Y2016.

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SIGNALS, ILLUMINATION, GROUND, GREEN STAR CLUSTER M125A1, RED STAR CLUSTER M158, WHITE STAR CLUSTER M159

A.1 SCOPE

A.1.1 Scope. This Appendix is a mandatory part of the specification. The information within is intended for compliance. This appendix presents the requirements and verification methods that are unique to Signals, Illumination, Ground, Green Star Cluster M125A1, Red Star Cluster M158, White Star Cluster M159, and should be used in combination with the rest of the requirements/verifications in the main part of this specification.

A.2 APPLICABLE DOCUMENTS

A.2.1 Other Government documents, Drawings, and publications. The following other Government documents, Drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (CCDC AC)

8797956	-	Illuminant Assembly, Green Star Cluster, M125A1
13010443	-	Tunnel for Light Intensity and Color Measurement
13058118	-	Illuminant Assemblies, Red Star Cluster (M158) and White Star Cluster (M159)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing

A.3 REQUIREMENTS

A.3.1 Required inspections.

A.3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1 and A4.1.

A.3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals and in accordance with 4.2 and A.4.2.

A.3.2 Components and assembly.

A.3.2.1 Illuminant assembly. The illuminant assembly shall comply with Drawing 8797956 for the Green Star Cluster M125A1 and Drawing 13058118 for the Red Star Cluster M158 and White Star Cluster M159.

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A.3.2.1.1 Candlepower, wavelength, purity, burntime. The candlepower, wavelength, purity and burntime when statically functioned shall comply with Drawing 8797956 for the Green Star Cluster M125A1 and Drawing 13058118 for the Red Star Cluster M158 and White Star Cluster M159. This is a static function test.

A.3.2.2 Assembly. The assembly shall be in accordance with Drawing 13061289.

A.3.2.3 Moisture content. The illuminant assembly shall have a moisture content that complies with Drawing 8797956 for the Green Star Cluster M125A1 and Drawing 13058118 for the Red Star Cluster M158 and White Star Cluster M159.

A.3.3 Functioning.

A.3.3.1 Ambient temperature (65°F to 75°F). The signal assemblies shall function and comply with the following:

a. Burning time of illuminant charges. All Cluster illuminant charges shall burn in 8.5 seconds ± 2 seconds.
b. Extinction interval. The interval between the third and fifth star extinguishing shall not exceed 4 seconds.
c. Incorrect color. The color emitted from the signal when functioned shall match the label on the signal.

A.3.3.2 Cold temperature (-70°F to -60°F). The signal assemblies shall function and comply with the following:

a. Burning time of illuminant charges. The requirements of A.3.3.1a.

b. Extinction interval. The requirements of A.3.3.1b.

A.3.4 Workmanship. The illuminant signal shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined and ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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A.4. VERIFICATION

TABLE A-I. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration (end item test) 3 - Examination 4 - Test		Classes of Verification A - First Article Inspection B – Conformance Inspection						
Description	Section A.3 Requirements	Verification Method				Verification Class		Section A.4 Verifications
		1	2	3	4	A	B	
First article	A.3.1.1	X	X	X	X	X		A.4.2
Conformance	A.3.1.2	X	X	X	X		X	A.4.3
Illuminant assembly	A.3.2.1	X	X	X	X	X	X	A.4.3.2.1
Assembly	A.3.2.2			X		X	X	A.4.3.2.2
Moisture content	A.3.2.3	X			X	X	X	A.4.3.3.1
Functioning – ambient	A.3.3.1		X		X	X	X	A.4.4
Functioning – Cold	A.3.3.2		X		X	X	X	A.4.4
Workmanship	A.3.4			X		X	X	A.4.3

A.4.1 Classification of verification. The verification requirements specified herein are classified as follows.

a. First Article Inspection (see A.4.2)
b. Conformance Inspection (see A.4.3)

A.4.2 First article inspection. When specified, a sample and its components and sub-assemblies shall be subjected to first article inspection in accordance with Table A-II.

A.4.2.1 Inspections to be performed. The first article inspection shall be 100% examination in accordance with Table A-II.

A.4.2.2 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table A-II are not conforming.

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TABLE A-II. First article inspection

Examination or Test	No. of Sample Units	Requirement Paragraph	Paragraph Reference/Inspection Method
Illuminant Assemblies			
Examination for defects	25	A.3.2.1	A.4.3.2.1
Static burn tests	20 each	A.3.2.1	A.4.3.2.1 & A.4.5.2
Assembly			
Examination for defects	25	A.3.2.2	A.4.3.2.2
Moisture content			
Examination for defects	3/	A3.2.3	A4.3.3.1
Functioning - Ambient	1/		
Burning time of illuminant charges	32	A.3.3	A.4.4
Extinction Interval	32	A.3.3	A.4.4
Functioning - Cold	2/		
Burning time of illuminant charges	16	A.3.3	A.4.4
Extinction Interval	16	A.3.3	A.4.4

Notes:
1/All Ambient functioning tests shall be measured in the same set performed in Table III, not independent sets.
2/ All Cold functioning tests shall be measured in the same set performed in Table III, not independent sets.
3/ Refer to A.4.3.3.1

A.4.3 Conformance inspection.

A.4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916. Refer to 4.3.1. In addition, each inspection lot of signal assemblies shall contain:

a. Composition ingredients from one lot.
b. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

A.4.3.2 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraph A.4.3.2.1 – A4.3.2.2, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.

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b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.

c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph “Acceptance by contractor-proposed provisions” of MIL-STD-1916).

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A.4.3.2.1	Illuminant assembly.			Drawing Number 8797956 & 13058118
				Next Higher Assembly 13061289
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Color designation incorrect	100% + VL-VII	A.3.2.1.1	Visual
Major				
101	Outside diameter, max	Level IV	A.3.2.1.1	Gage
102	Quickmatch missing or not partially embedded in priming charge, when applicable	Level IV	A.3.2.1.1	Visual/Manual
103	Candlepower, wavelength, purity 1/	2/	A.3.2.1.1	A.4.5.2
Minor				
201	Assembly damaged	Level II	A.3.2.1.1	Visual
202	Marking missing or incorrect	Level II	A.3.2.1.1	Visual
203	Priming charge on exposed portion of quickmatch (where applicable)	Level II	A.3.2.1.1	Visual
204	Evidence of poor workmanship	Level II	A.3.4	Visual

Notes:
1/ This test is for acceptance of illuminant assembly lots prior to use in production.
2/ Twenty (20) samples/lot or batch - same sample for both tests. Candlepower,: acc 1, rej 2; wavelength, purity: acc 0, rej 1; gross incorrect color differentiation (green vs red) is a critical

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A.4.3.2.2	Assembly (signal body & delay assembly after assembly of tail, propellant, exhaust plate, washer & discs, and illuminant assemblies, bolt and backing washer).			Drawing Number 13061289
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Illuminant assembly missing	Level IV	A.3.2.2	Visual
102	Washer disc assembly missing	Level IV	A.3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	A.3.4	Visual

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A.4.3.3 Production testing. Unless otherwise specified herein, the following tests shall be performed using MIL-STD-1916 for sampling plan requirements.

A.4.3.3.1 Moisture content. Major Defect. (See Table A-III)

TABLE A-III. Moisture content.

Material	Drawing	Inspection Method Paragraph
Illuminant composition - Green Star	8797956	A.4.5.1
- White Star	13058118	A.4.5.1
- Red Star	13058118	A.4.5.1

Adequate controls shall be provided to insure that the materials comply with the requirements. For verification, a sample shall be selected and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excessive moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in A.4.5.1. (See 6.8)

A.4.4 Functioning. The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table A-IV and Table VI when tested as specified in A.4.4.1 and A.4.4.2.

TABLE A-IV. Functioning defects.

Examination or Test	Category	Requirement Paragraph	Inspection Method
At Ambient and Cold			
Burning time incorrect	Major	A.3.3.1a	Gage
Extinction interval incorrect	Major	A.3.3.1b	Gage

A.4.4.1 Ambient temperature (65°F to 75°F). The packaged signals shall be conditioned with the requirements as classified in paragraph 4.4.2. The functioning performance shall be observed for compliance with the requirements of A.4.4.1.1.

A.4.4.1.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table A-IV and Table V, one hundred (100) assemblies shall be selected from each lot for testing and

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conditioned in accordance with A.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table A-IV, Table V, and Table VI.

A.4.4.1.2 Reduced sampling plan. After three consecutive lots have complied with the criteria of A.4.4.1.1, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with A.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table A-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, A.4.4.1, and A.4.4.2.

A.4.4.2 Cold temperature (-70°F to -60°F). The packed signals shall be conditioned for a minimum of 16 hours at -70°F to -60°F. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples must be fired within three minutes of removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements of A.4.4.2.1.

A.4.4.2.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table A-IV, Table V, and Table VI, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with A.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table A-IV, Table V, and Table VI. A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.

A.4.4.2.2 Reduced sampling plan. After three (3) consecutive lots have complied with the criteria of A.4.4.2.1, thirty-two (32) assemblies shall be selected from each lot for testing and conditioned in accordance with A.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table A-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, A.4.4.1, and A.4.4.2. A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.

A.4.5 Test methods and procedures.

A.4.5.1 Illuminant composition moisture content. The Karl Fischer Method 101.2.1 as stated in MIL-STD-1234 shall be used. A fifty gram sample plus or minus 0.1 gram shall be added to a 500 milliliter (ml) volumetric flask containing approximately 300 to 400 ml of methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 ml volumetric flask shall be filled up to the 500 ml mark with methanol and swirled again.

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A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml aliquot of the clear supernatant liquid shall be withdrawn and immediately transformed to the standard titration vessel containing approximately 100 ml of methanol which has just been titrated to the preliminary end point as described in MIL-STD-1234. The final end point shall be reached in 3.5 minutes in the manner described in MIL-STD-1234. A 50 ml aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

$$\%\, Water = \frac{100M * ((VR - S) - (V'R - S'))}{W}$$

Where:

M = grams of water per ml of standard water in methanol solution
V = ml of Karl Fischer reagent added to the sample
V' = ml of Karl Fischer reagent added to the blank
R = ml of standard water in methanol solution per ml of Karl Fischer reagent
S = ml of standard water in methanol solution for titration of sample
S' = ml of standard water in methanol solution with back titration of blank
W = weight of sample in grams

A.4.5.2 Candlepower, wavelength, purity and burntime. The illuminant assemblies shall be tested for candlepower, wavelength, purity and burntime in accordance with the procedure and equipment specified in Drawing 13010443. If any sample fails to comply with the applicable requirement, the lot shall be rejected. Illuminant assemblies from each mix, batch or sublot making up the inspection lot shall be uniformly represented in the above sample.

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SIGNALS, ILLUMINATION, GROUND, RED STAR PARACHUTE M126A1, WHITE STAR PARACHUTE M127A1, AND GREEN STAR PARACHUTE M195

B.1 SCOPE

B.1.1 Scope. This Appendix is a mandatory part of the specification. The information within is intended for compliance. This appendix presents the requirements and verification methods that are unique to Signals, Illumination, Ground, Red Star Parachute M126A1, White Star Parachute M127A1, and Green Star Parachute M195 and shall be used in combination with the rest of the requirements/verifications in the main part of this specification.

B.2 APPLICABLE DOCUMENTS

B.2.1 Other Government documents, Drawings, and publications. The following other Government documents, Drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (CCDC AC)

8797991	-	Parachute Assembly
9255113	-	Illuminant Assembly, Green Star, M195
13010443	-	Tunnel for Light Intensity and Color Measurement
13058117	-	Illuminant Assembly, Red Star, M126A1
13058119	-	Illuminant Assembly, Mod. White Star M127A1

B.3 REQUIREMENTS

B.3.1 Required inspections.

B.3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1 and B.4.1.

B.3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals and in accordance with 4.2 and B.4.2.

B.3.2 Components and assembly.

B.3.2.1 Illuminant assembly. The illuminant assembly shall comply with Drawing 9255113 for the Green Star Parachute M195, Drawing 13058119 for the White Star Parachute M127A1, and Drawing 13058117 for the Red Star Parachute M126A1.

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B.3.2.1.1 Candlepower, wavelength, purity, burntime. The candlepower, wavelength, purity and burntime shall comply with Drawing 9255113 for the Green Star Parachute M195, Drawing 13058119 for the White Star Parachute M127A1, and Drawing 13058117 for the Red Star Parachute M126A1. This is a static function test.

B.3.2.2 Parachute assembly. The parachute shall be assembled in accordance with Drawing 8797991.

B.3.2.3 Assembly. The assembly shall be in accordance with Drawing 13061290.

B.3.2.4 Moisture content. The illuminant assembly shall have a moisture content that shall comply with Drawing 9255113 for the Green Star Parachute M195, Drawing 13058119 for the White Star Parachute M127A1, and Drawing 13058117 for the Red Star Parachute M1126A1.

B.3.3 Functioning.

B.3.3.1 Ambient temperature (65°F to 75°F). The signal assemblies shall function and comply with the following:

a. Burn time. The illuminant assembly shall burn for not less than fifty (50) seconds in the M126A1 and the M195 assembles, and for not less than twenty-five (25) seconds in the M127A1 assembly.
b. Parachute failure. The parachute shall remain fully open after not more than five (5) seconds from the time of ejection, and shall not separate from the assembly.
c. Incorrect color. The color emitted from the signal when functioned shall match the label on the signal.

B.3.3.2 Cold temperature (-70°F to -60°F). The signal assemblies shall function and comply with the following:

a. Burn time. The requirements of B.3.3.1a.
b. Parachute Separation. The parachute shall not separate from the illuminant assembly (Parachute opening delay shall be recorded for informational purposes as well as whether the chute remains partially open).

B.3.4 Workmanship. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined and ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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B.4. VERIFICATION

TABLE B-I. Requirement/verification cross reference matrix.

Method of Verification 1 - Analysis 2 - Demonstration (end item test) 3 - Examination 4 - Test		Classes of Verification A - First article inspection B – Conformance inspection						
Description	Section B.3 Requirements	Verification Method				Verification Class		Section B.4 Verifications
		1	2	3	4	A	B	
First Article	B.3.1.1	X	X	X	X	X		B.4.2
Conformance	B.3.1.2	X	X	X	X		X	B.4.3
Illuminant assembly	B.3.2.1	X	X	X	X	X	X	B.4.3.2.1
Parachute assembly	B.3.2.2	X	X	X	X	X	X	B.4.3.2.2
Assembly	B.3.2.3			X		X	X	B.4.3.2.3
Moisture content	B.3.2.4	X			X	X	X	B.4.3.3.1
Functioning – Ambient	B.3.3.1		X		X	X	X	B.4.4
Functioning – Cold	B.3.3.2		X		X	X	X	B.4.4
Workmanship	B.3.4			X		X	X	B.4.3

B.4.1 Classification of verification. The verification requirements specified herein are classified as follows.

a. First Article Inspection (see B.4.2)
b. Conformance Inspection (see B.4.3)

B.4.2 First article inspection. When specified, a sample and its components and sub assemblies shall be subjected to first article inspection in accordance with Table B-II.

B.4.2.1 Inspections to be performed. The first article inspection shall be 100% examination of all Drawings characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed and the tests specified in Table B-II.

B.4.2.2 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table B-II are not conforming.

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TABLE B-II. First Article Inspection

Examination or Test	No. of Sample Units	Requirement Paragraph	Paragraph Reference/Inspection Method
Illuminant assemblies			
Examination for defects	25	B.3.2.1	B.4.3.2.1
Static burn tests	20 each	B.3.2.1.1	B.4.3.2.1 & B.4.5.2
Parachute assemblies			
Examination for defects	25	B.3.2.2	B.4.3.2.2
Assembly			
Examination for defects	25	B.3.2.3	B.4.3.2.3
Moisture content			
Examination for defects	3/	B.3.2.4	B.4.3.3.1
Functioning - Ambient	1/		
Burning time of illuminant charges	32	B.3.3.1a	B.4.4
Parachute opening	32	B.3.3.1b	B.4.4
Functioning - Cold	2/		
Burning time of illuminant charge	16	B.3.3.2a	B.4.4
Parachute opening	16	B.3.3.2b	B.4.4

Notes:
1/ All Ambient functioning tests shall be measured in the same set performed in Table III, not independent sets.
2/ All Cold functioning tests shall be measured in the same set performed in Table III, not independent sets.
3/ Refer to B.4.3.3.1

B.4.3 Conformance inspection.

B.4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916. Refer to 4.3.1. In addition, each inspection lot of signal assemblies shall contain:

a. Composition ingredients from one lot.
b. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

B.4.3.2 Inspection requirements by classification of characteristics.

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a. For the conformance inspection paragraph A.4.3.2.1 – A4.3.2.2, the definitions of critical, major and minor defects are provided in paragraph "Definitions" of MIL-STD-1916.

b. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916.

c. Alternative conformance acceptance. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

B.4.3.2.1	Illuminant assembly.			Drawing Number 9255113, 13058119, & 13058117
				Next Higher Assembly 13061290
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Color designation incorrect	100% + VL-VII	B.3.2.1.1	Visual
Major				
101	Outside diameter, max	Level IV	B.3.2.1.1	Gage
102	Quickmatch missing or not partially embedded in priming charge, when applicable	Level IV	B.3.2.1.1	Visual/Manual
103	Candlepower, wavelength, purity 1/	2/	B.3.2.1.1	B.4.5.2
Minor				
201	Assembly damaged	Level II	B.3.2.1	Visual
202	Marking missing or incorrect	Level II	B.3.2.1	Visual
203	Priming charge on exposed portion of quickmatch (where applicable)	Level II	B.3.2.1	Visual
204	Evidence of poor workmanship	Level II	B.3.4	Visual

Notes:
1/ This test is for acceptance of illuminant assembly lots prior to use in production.
2/ Twenty (20) samples/lot or batch - same sample for both tests. Candlepower: acc 1, rej 2; wavelength, purity: acc 0, rej 1

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B.4.3.2.2	Parachute assembly.			Drawing Number 8797991
				Next Higher Assembly 13061290
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Parachute torn, contains holes or is improperly tied to suspension cord or improperly attached to chain bead assembly, when applicable	Level IV	B.3.2.2	Visual
102	Shrouds twisted	Level IV	B.3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	B.3.4	Visual

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B.4.3.2.3	Assembly (signal body and delay assembly after assembly of tail assembly, propellant assembly, exhaust plate and bolt, washer disc assembly and illuminant assembly, but prior to inserting parachute in body).			Drawing Number 13061290
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Suspension cord insecurely tied to illuminant assembly	Level IV	B.3.2.3	Visual
102	Component missing or damaged to the extent that function may be impaired or chain bead assembly improperly fastened to illuminant assembly	Level IV	B.3.2.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	B.3.4	Visual

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B.4.3.3 Production testing. Unless otherwise specified herein, the following tests shall be performed using MIL-STD-1916 for sampling plan requirements.

B.4.3.3.1 Moisture content. Major Defect. (See Table B-III)

TABLE B-III. Moisture Content.

Material	Drawing	Inspection Method Paragraph
Illuminant composition - Green Star Parachute	9255113	B.4.5.1.1
- White Star Parachute	13058119	B.4.5.1.1
- Red Star Parachute	13058117	B.4.5.1.1
Parachute Assembly	8797991	B.4.5.1.2

Adequate controls shall be provided to insure that the materials comply with the requirement. For verification, a sample shall be selected and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excessive moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in B.4.5.1.

B.4.4 Functioning. The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table B-IV and Table VI when tested as specified in B.4.4.1 and B.4.4.2.

TABLE B-IV. Functioning Defects.

Examination or Test	Category	Requirement Paragraph	Inspection Method
At both Ambient and Cold			
Burning time incorrect	Major	B.3.3.1a	Gage
Cold			
Parachute separation	Major	B.3.3.2b	Gage/Visual
Ambient			
Parachute failure	Major	B.3.3.1b	Gage/Visual

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B.4.4.1 Ambient temperature (65°F to 75°F). The packed signals shall be conditioned for a minimum of 16 hours at 65°F to 75°F. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples must be fired within three minutes of removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements of B.4.4.1.1.

B.4.4.1.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table B-IV and Table V, one hundred (100) assemblies shall be selected from each lot for testing and conditioned in accordance with B.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table B-IV, Table V, and Table VI.

B.4.4.1.2 Reduced sampling plan. After three consecutive lots have complied with the criteria of B.4.4.1.1, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with B.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table B-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, B.4.4.1, and B.4.4.2.

B.4.4.2 Cold temperature (-70°F to -60°F). The packed signals shall be conditioned for a minimum of 16 hours at -70°F to -60°F. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples must be fired within three minutes of removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements of B.4.4.2.1.

B.4.4.2.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table B-IV, Table V, and Table VI, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with B.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table B-IV, Table V, and Table VI. A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.

B.4.4.2.2 Reduced sampling plan. After three (3) consecutive lots have complied with the criteria of B.4.4.2.1, thirty-two (32) assemblies shall be selected from each lot for testing and conditioned in accordance with B.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table B-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, B.4.4.1, and B.4.4.2. A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.

B.4.5 Test methods and procedures.

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B.4.5.1 Moisture content.

B.4.5.1.1 Illuminant composition moisture content. The Karl Fischer Method 101.2.1 as stated in MIL-STD-1234 shall be used. A fifty gram sample plus or minus 0.1 gram shall be added to a 500 milliliter (ml) volumetric flask containing approximately 300 to 400 ml of methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 ml volumetric flask shall be filled up to the 500 ml mark with methanol and swirled again. A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml aliquot of the clear supernatant liquid shall be withdrawn and immediately transformed to the standard titration vessel containing approximately 100 ml of methanol which has just been titrated to the preliminary end point as described in MIL-STD-1234. The final end point shall be reached in 3.5 minutes in the manner described in MIL-STD-1234. A 50 ml aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

Percent water equal (VR-S) – (V'R-S')
Multiply the result by 100F and divide by W
Where:

F =	grams of water per ml of standard water in methanol solution
V =	ml of Karl Fischer reagent added to the sample
V' =	ml of Karl Fischer reagent added to the blank
R =	ml of standard water in methanol solution per ml of Karl Fischer reagent
S =	ml of standard water in methanol solution for titration of sample
S' =	ml of standard water in methanol solution with back titration of blank
W =	weight of sample in grams

B.4.5.1.2 Parachute cloth. An accurately weighed portion of at least 10 grams of the material weighed to the nearest milligram shall be transferred to a tared weighing dish and replaced in an oven and heated at a temperature of 100 degrees plus or minus 2 degrees Centigrade (C) for 2 hours. The dish and contents shall be removed from the oven and placed in a desiccator, cooled, and weighed. The difference in weight shall be calculated to percentage of moisture in the sample.

B.4.5.2 Candlepower, wavelength, purity, and burntime. The illuminant assemblies shall be tested for candlepower, wavelength, purity and burntime in accordance with the procedure and equipment specified in Drawing 13010443. If any sample fails to comply with the applicable requirement, the lot shall be rejected. Illuminant assemblies from each mix, batch or sublot making up the inspection lot shall be uniformly represented in the above sample.

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SIGNALS, SMOKE, GROUND, HAND HELD PARACHUTE, GREEN SMOKE, M128A1, RED SMOKE, M129A1, AND YELLOW SMOKE, M194

C.1 SCOPE

C.1.1 Scope. This Appendix is a mandatory part of the specification. The information within is intended for compliance. This appendix presents the requirements and verification methods that are unique to Signals, Smoke, Ground, Hand Held Parachute, Green Smoke, M128A1, Red Smoke, M129A1, and Yellow Smoke, M194 and shall be used in combination with the rest of the requirements/verifications in the main part of this specification.

C.2 APPLICABLE DOCUMENTS

C.2.1 Other Government documents, Drawings, and publications. The following other 6Government documents, Drawings and publications form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (CCDC AC)

8797991	-	Parachute Assembly
8797998	-	Smoke Assembly (M128A1 and M129A1)
9255783	-	Yellow Smoke Assembly

C.3 REQUIREMENTS

C.3.1 Required inspections.

C.3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1 and C.4.1.

C.3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with Drawing 13061289 for cluster signals, Drawing 13061290 for parachute signals, Drawing 8797996 for green and red smoke signals, and Drawing 9255782 for yellow smoke signals and in accordance with 4.2 and C.4.2.

C.3.2 Components and assembly.

C.3.2.1 Smoke assembly. The smoke assembly shall comply with Drawing 8797998 for the Green Smoke Parachute, M128A1 and Red Smoke Parachute, M129A1, and Drawing 9255783 for the Yellow Smoke Parachute.

C.3.2.2 Parachute assembly. The parachute shall be assembled in accordance with Drawing 8797991.

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C.3.2.3 Assembly. The assembly shall be in accordance with Drawing 8797996 for the Green Smoke Parachute, M128A1 and Red Smoke Parachute, M129A1, and Drawing 9255782 for the Yellow Smoke Parachute.

C.3.2.4 Moisture content. The illuminant assembly shall have a moisture content that shall comply with Drawing 8797998 for the Green Smoke Parachute, M128A1 and Red Smoke Parachute, M129A1, and Drawing 9255783 for the Yellow Smoke Parachute.

C.3.3 Functioning.

C.3.3.1 Ambient temperature (65°F to 75°F). The signal assemblies shall function and comply with the following:

a. Ambient burn time. The burning time of the smoke assembly shall not be less than 6.0 seconds or more than 18 seconds for the M128, and not less than 18.0 seconds or more than 40 seconds for the M129, and not less than 9.0 seconds or more than 18 seconds for the M194.
b. Parachute failure. The parachute shall remain fully open after not more than five (5) seconds from the time of ejection, and shall not separate from the assembly.
c. Smoke emission. The smoke assembly shall start to emit smoke not more than 4 seconds after ejection from the signal.

C.3.3.2 Cold temperature (-70°F to -60°F). The signal assemblies shall function and comply with the following:

a. Cold burn time. The burning time of the smoke assembly shall not be less than 6.0 seconds or more than 22 seconds for the M128, and not less than 20.0 seconds or more than 60 seconds for the M129, and not less than 9.0 seconds or more than 18 seconds for the M194.
b. Parachute separation. The parachute shall not separate from the illuminant assembly (Parachute opening delay shall be recorded for informational purposes as well as whether the chute remains partially open).
c. Smoke emission. The requirements of C.3.3.1c

C.3.4 Workmanship. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined and ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

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C.4. VERIFICATION

TABLE C-I. Requirement/verification cross reference matrix

Method of Verification 1 - Analysis 2 - Demonstration (end item test) 3 - Examination 4 - Test		Classes of Verification A - First article inspection B – Conformance inspection						
Description	Section C.3 Requirements	Verification Method				Verification Class		Section C.4 Verifications
		1	2	3	4	A	B	
First article	C.3.1	X	X	X	X	X		C.4.2
Conformance	C.3.1.2	X	X	X	X		X	C.4.3
Smoke assembly	C.3.2.1	X	X	X	X	X	X	C.4.3.2.1
Parachute assembly	C.3.2.2	X	X	X	X	X	X	C.4.3.2.2
Assembly	C.3.2.3			X		X	X	C.4.3.2.3
Moisture content	C.3.2.4	X			X	X	X	C.4.3.3.1
Functioning – Ambient	C.3.3.1		X		X	X	X	C.4.4
Functioning – Cold	C.3.3.2		X		X	X	X	C.4.4
Workmanship	C.3.4			X		X	X	C.4.3

C.4.1 Classification of verification. The verification requirements specified herein are classified as follows.

a. First Article Inspection (see C.4.2)
b. Conformance Inspection (see C.4.3)

C.4.2 First article inspection. When specified, a sample and its components and sub assemblies shall be subjected to first article inspection in accordance with Table C-II.

C.4.2.1 Inspections to be performed. The first article inspection shall be 100% examination of all Drawings characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed and the tests specified in Table C-II.

C.4.2.2 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table C-II are not conforming.

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TABLE C-II. First Article Inspection

Examination or Test	No. of Sample Units	Requirement Paragraph	Paragraph Reference/Inspection Method
Smoke assemblies			
Examination for defects	25	C.3.2.1	C.4.3.2.1
Parachute assembly			
Examination for defects	25	C.3.2.2	C.4.3.2.2
Assembly			
Examination for defects	25	C.3.2.3	C.4.3.2.3
Moisture content			
Examination for defects	25	C.3.2.4	C.4.3.3.1 & C.4.5.1
Functioning - Ambient	1/		
Ambient burn time	32	C.3.3.1a	C.4.4.1
Parachute failure	32	C.3.3.1b	C.4.4.1
Smoke emission	32	C3.3.1c	C.4.4.1
Functioning - Cold	2/		
Burning time of illuminant charge	16	C.3.3.2a	C.4.4.2
Parachute Opening	16	C.3.3.2b	C.4.4.2
Smoke emission	16	C.3.3.2c	C.4.4.2

Notes:
1/ All Ambient functioning tests shall be measured in the same set performed in Table III, not independent sets.
2/ All Cold functioning tests shall be measured in the same set performed in Table III, not independent sets.

C.4.3 Conformance inspection.

C.4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-STD-1916. Refer to 4.3.1. In addition, each inspection lot of signal assemblies shall contain:

a. Composition ingredients from one lot.
b. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

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C.4.3.2.1	Smoke assembly.			Drawing Number 8797998/9255783
				Next Higher Assembly 8797996/9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical				
1	Color designation incorrect	100% + VL-VII	C.3.2.1	Visual
Major				
101	Outside diameter, max	Level IV	C.3.2.1	Gage
102	Any component missing	Level IV	C.3.2.1	Visual
103	Wrapper overlaps holes	Level IV	C.3.2.1	Visual
104	Quickmatch missing or not partially embedded in priming charge, when applicable	Level IV	C.3.2.1	Visual/Manual
Minor				
201	Total length, max.	Level II	C.3.2.1	Gage
202	Marking missing or incorrect	Level II	C.3.2.1	Visual
203	Assembly damaged	Level II	C.3.2.1	Visual
204	Ends of quickmatch overlapping the 360 degree crimp	Level II	B.3.2.1	Visual
205	Evidence of poor workmanship	Level II	C.3.2.1	Visual

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C.4.3.2.2	Parachute assembly.			Drawing Number 8797991
				Next Higher Assembly 8797996/9255782
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Parachute torn, contains holes or is improperly tied to suspension cord or improperly attached to chain bead assembly, when applicable	Level IV	C.3.2.2	Visual
102	Shrouds twisted	Level IV	C.3.2.2	Visual
Minor				
201	Evidence of poor workmanship	Level II	C.3.4	Visual

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C.4.3.2.3	Assembly (signal body and delay assembly after assembly of tail assembly, propellant assembly, exhaust plate and bolt, washer disc assembly and smoke assembly, but prior to inserting parachute in body).			Drawing Number 8797996/9255782
				Next Higher Assembly - - -
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Parachute improperly tied to swivel	Level IV	C.3.2.3	Visual
102	Component missing or damaged to the extent that function may be impaired or chain bead assembly improperly fastened to illuminant assembly	Level IV	C.3.2.3	Visual
Minor				
201	Evidence of poor workmanship	Level II	C.3.4	Visual

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C.4.3.3 Production testing. Unless otherwise specified herein, the following tests shall be performed using MIL-STD-1916 for sampling plan requirements.

C.4.3.3.1 Moisture content. Major Defect. (See Table B-III)

TABLE C-III. Moisture Content.

Material	Drawing	Inspection Method Paragraph
Smoke composition - Green Star Parachute	8797998	C.4.5.1.1
- White Star Parachute	9255783	C.4.5.1.1
- Red Star Parachute	8797998	C.4.5.1.1
Parachute Assembly	8797991	C.4.5.1.2

Adequate controls shall be provided to insure that the materials comply with the requirement. For verification, a sample shall be selected and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of the sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excessive moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in C.4.5.1. (See 6.10)

C.4.4 Functioning. The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table C-IV and Table VI when tested as specified in C.4.4.1 and C.4.4.2.

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TABLE C-IV. Functioning Defects.

Examination or Test	Category	Requirement Paragraph	Inspection Method
At both Ambient and Cold			
Smoke emission time over max	Major	C.3.3.1c & C3.3.2c	Gage
Cold			
Cold burn time	Major	C.3.3.2a	Gage
Parachute separation	Major	C.3.3.2b	Gage/Visual
Ambient			
Ambient burn time	Major	C.3.3.1a	Gage
Parachute failure	Major	C.3.3.1b	Gage/Visual

C.4.4.1 Ambient temperature (65°F to 75°F). The packed signals shall be conditioned for a minimum of 16 hours at 65°F to 75°F. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples must be fired within three minutes of removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements of C.4.4.1.1.

C.4.4.1.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table C-IV and Table V, one hundred (100) assemblies shall be selected from each lot for testing and conditioned in accordance with C.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table C-IV, Table V, and Table VI.

C.4.4.1.2 Reduced sampling plan. After three consecutive lots have complied with the criteria of B.4.4.1.1, fifty (50) assemblies shall be selected from each lot for testing and conditioned in accordance with C.4.4.1. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table C-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, C.4.4.1, and C.4.4.2.

C.4.4.2 Cold temperature (-70°F to -60°F). The packed signals shall be conditioned for a minimum of 16 hours at -70°F to -60°F. The packed signal shall be taken from the conditioning chamber, removed from the plastic container and fired vertically from a fixture. Samples must be fired within three minutes of removal from the conditioning chamber. The functioning performance shall be observed for compliance with the requirements of C.4.4.2.1.

C.4.4.2.1 Initial sampling plan. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified in Table C-IV, Table V, and Table VI, fifty (50) assemblies shall be selected from each lot for testing and

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conditioned in accordance with C.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table C-IV, Table V, and Table VI. A smoke assembly which ignites and burns normally but subsequently fails to meet minimum burning time requirements due to ground impact will not be classified defective. Firing records will be annotated accordingly.

C.4.4.2.2 Reduced sampling plan. After three (3) consecutive lots have complied with the criteria of B.4.4.2.1, thirty-two (32) assemblies shall be selected from each lot for testing and conditioned in accordance with C.4.4.2. The lot shall be rejected if it fails to meet the acceptance criteria specified in Table C-IV, Table V, and Table VI. If the lot is rejected, or a new First Article is required, the next lot produced shall revert to the initial sampling plan for each functioning temperature in accordance with the requirements of 4.4.1, 4.4.2, 4.4.3, C.4.4.1, and C.4.4.2. A smoke assembly which ignites and burns normally but subsequently fails to meet minimum burning time requirements due to ground impact will not be classified defective. Firing records will be annotated accordingly.

C.4.5 Test methods and procedures.

C.4.5.1 Moisture content.

C.4.5.1.1 Smoke composition moisture content. The moisture content of the smoke compositions shall be determined in accordance with the method specified in MIL-STD-1234 method 101.1.1 except that the test drying time shall be 4 hours and vacuum shall be applied.

C.4.5.1.2 Parachute cloth. An accurately weighed portion of at least 10 grams of the material weighed to the nearest milligram shall be transferred to a tared weighing dish and replaced in an oven and heated at a temperature of 100 degrees plus or minus 2 degrees Centigrade (C) for 2 hours. The dish and contents shall be removed from the oven and placed in a desiccator, cooled, and weighed. The difference in weight shall be calculated to percentage of moisture in the sample.

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CONCLUDING MATERIAL

Custodian:
Army-AR
Navy - OS
Air Force - 170

Preparing Activity:
Army-AR
(Project 1370-2021-002)

Review Activities:
Army – AV, MI
Navy - AS

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